

FinSentinel — STAGE04 User Manual

FinSentinel — Financial Crime Risk Assessment Workbench

AI prepares. Rules calculate. Humans decide.

Field	Value
Stage	STAGE04 — Testing (validation, quality assurance, test automation)
Date	2026-09-30
Audience	Product Owners, FCRM Analysts, Risk Committee members, FCRM Administrators, and the judging panel
How it was produced	Every screenshot comes from one scripted walkthrough of the real application in Chromium (<code>npm run manual:capture</code> , <code>tests/manual/capture.spec.ts</code>). The diagrams are generated from the workflow and RBAC source (<code>scripts/manual/diagrams.ts</code>). Nothing is mocked up by hand.
Data	Synthetic only. Four seeded simulation users, two seeded demo cases (RA-1001, RA-1002), sample documents per change type.
Companion documents	<code>STAGE04_Testing_Report.md</code> (what was tested), <code>STAGE02_Design_UX.md</code> (screen design), <code>STAGE02_Design_Architecture.md</code> (behaviour)

1. What FinSentinel does

FinSentinel prepares evidence-backed financial-crime risk assessments for new products, features, vendors, geographies, channels, customer segments, processes and transaction changes. A code-orchestrated AI pipeline reads the submitted documents, extracts facts with their sources, retrieves the applicable policy sections and drafts an assessment in which every claim cites a quote. A deterministic risk engine calculates inherent and residual risk from a versioned risk model. People then do the parts that matter: the analyst challenges and corrects, the committee decides, and every action lands in an append-only, hash-chained audit trail.

Three rules shape every screen in this manual:

- **AI prepares.** AI output is labelled *AI prepared* and is advisory. Unsupported claims are flagged, never hidden (FR-ASM-02, FR-ASM-03).
- **Rules calculate.** Risk scores come from the risk engine, not the model. Controls reduce risk but never eliminate it (FR-RSK-01, FR-RSK-08).
- **Humans decide.** Only a Risk Committee user can record a decision, and only after the analyst has finalized (FR-WF-07, FR-WF-09). An AI failure leaves the case pending; it never approves or rejects (FR-FAIL-04).

2. Getting started

2.1 Install and start

On Windows with Node 24, run the installer once from the repository root. It checks the toolchain, sets the proxy trust variable, creates `.env` with a generated secret, installs dependencies, migrates and seeds the database and builds the web app.

```
powershell -NoProfile -ExecutionPolicy Bypass -File .\scripts\install.ps1
npm start # http://localhost:3000
```

The default configuration uses the mock LLM gateway, which replays recorded fixtures, so no API key is needed. Set `LLM_GATEWAY=anthropic` and `ANTHROPIC_API_KEY` in `.env` to use the real model. The rest of this manual is identical either way.

2.2 Sign in

Sign-in is **simulation mode**: pick a role tile, no password. Each tile is a seeded synthetic user. The header always shows the active role badge and user, and the "Synthetic data · simulation sign-in" pill reminds you that nothing here is real.

Tile	Role	What they do
Pat Owner	Product Owner	Creates cases, attaches documents, answers information requests, follows the decision
Kim Analyst	FCRM Analyst	Reviews the AI's work, clears blockers, sets controls, overrides with rationale, finalizes
Lee Committee	Risk Committee	Reviews the packet and records the decision
Raj Admin	FCRM Administrator	Publishes risk model versions, watches the Quality Center and Ops

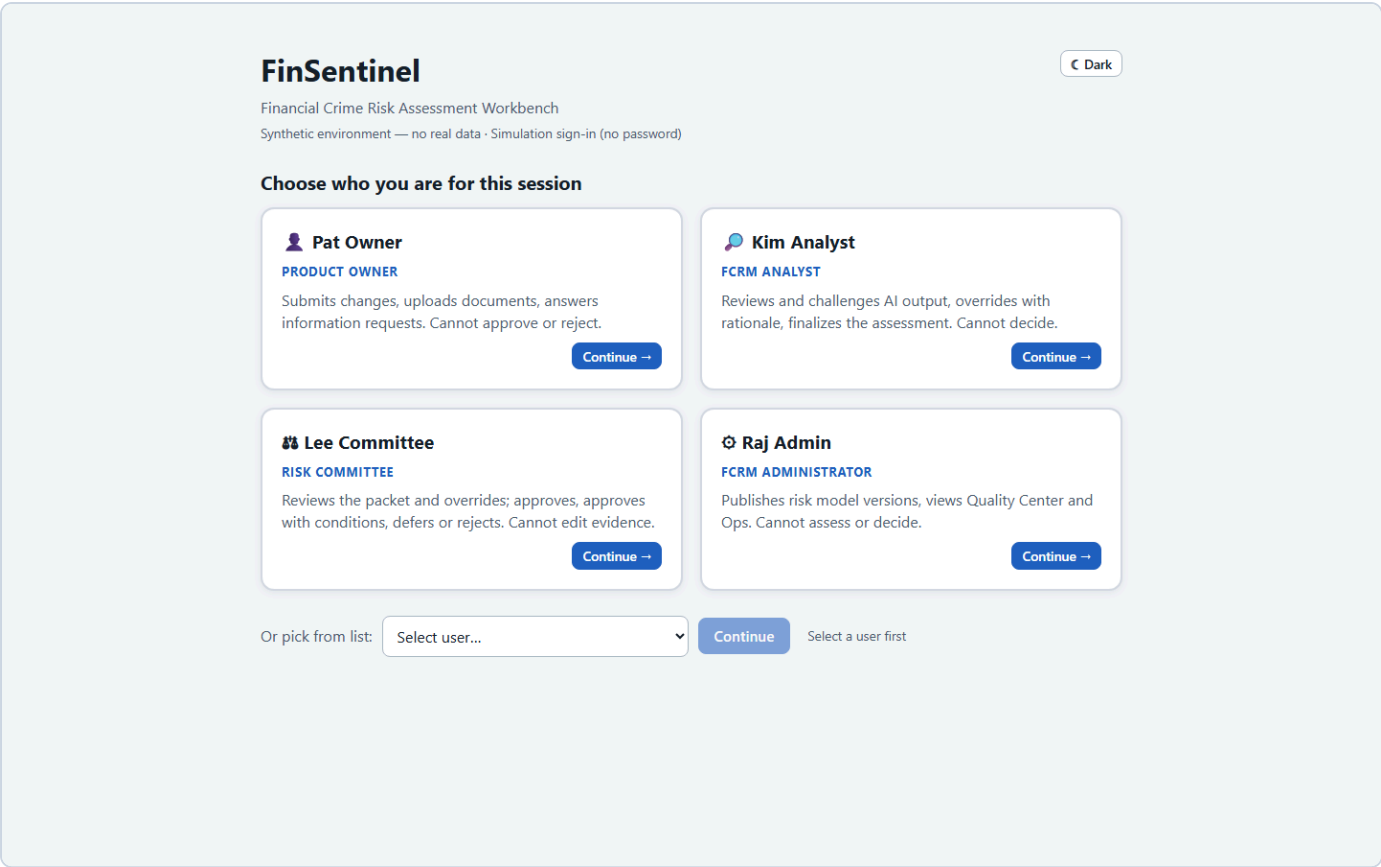


Figure 2.1 — Sign-in with role tiles (UX §4.1). The theme toggle in the header switches between light and dark and the choice persists across reloads.

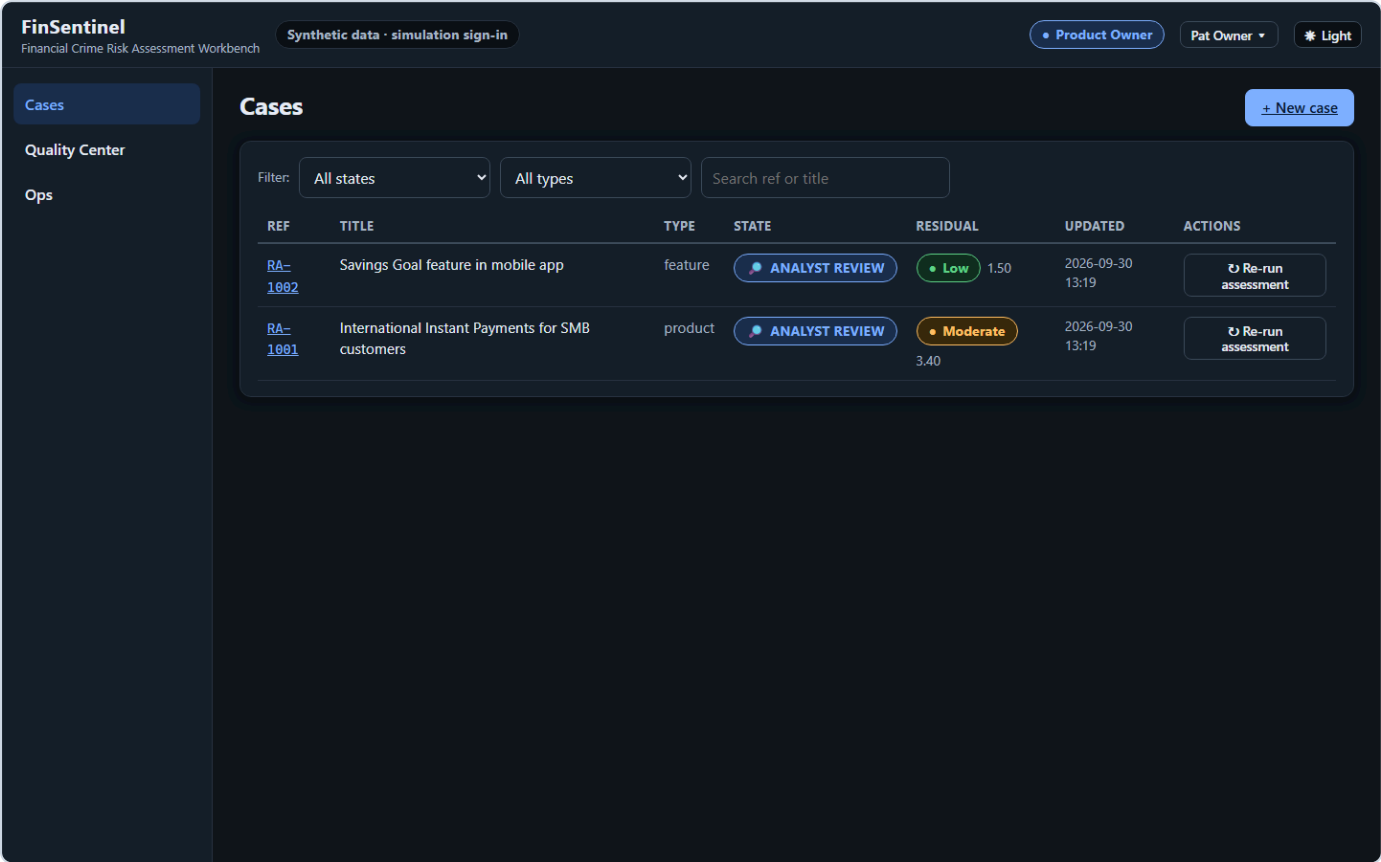


Figure 2.2 — The same application in dark theme (UX §4.1a).

2.3 Navigation

The left rail has **Cases**, **Quality Center** and **Ops** for every role, and **Admin** for the administrator only. A case opens as a workspace with seven tabs: Overview, Facts, Evidence, Assessment, Risk, Committee and History. Tabs show a count badge when they contain something that needs attention. Actions a role cannot perform are absent from its screens rather than greyed out (UX §7).

3. The workflow at a glance

3.1 Case lifecycle

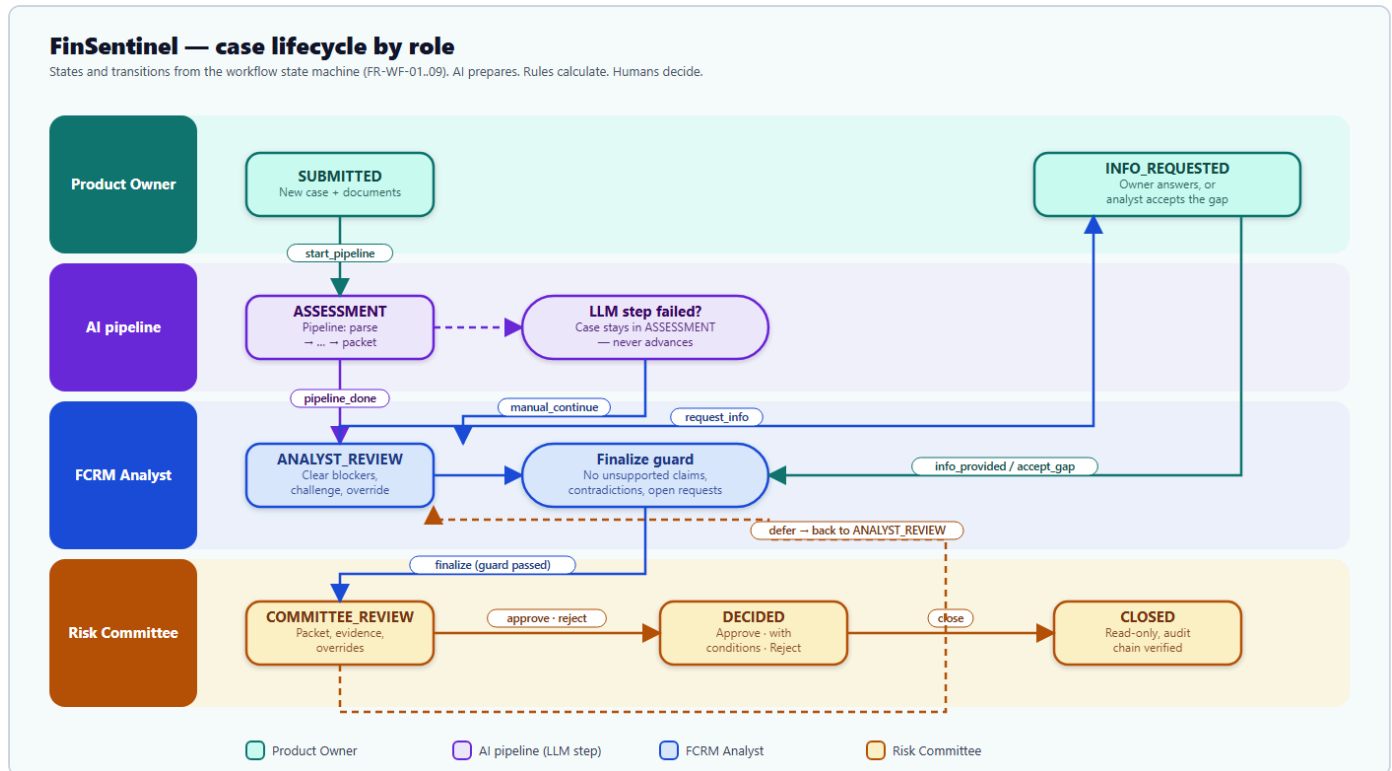


Figure 3.1 — Case states and transitions, one swim lane per actor. Generated from the workflow state machine in `src/domain/workflow.ts` (FR-WF-01..09).

State	Meaning	Who moves it on
SUBMITTED	Case created with documents	Product Owner or Analyst starts the AI review
ASSESSMENT	The pipeline is running	The pipeline finishes, or the analyst uses <i>manual continue</i> after a failed LLM step
ANALYST_REVIEW	Analyst clears blockers, challenges and overrides	Analyst requests information or finalizes
INFO_REQUESTED	Waiting on the Product Owner	Owner answers, or the analyst accepts the gap with rationale
COMMITTEE_REVIEW	Packet is ready for decision	Committee approves, approves with conditions, rejects, or defers back to the analyst
DECIDED	Decision recorded with rationale and conditions	Committee closes the case
CLOSED	Read-only; audit chain verifiable	Nobody. Closed cases cannot change.

An invalid transition is rejected by the server (FR-WF-02). A case cannot skip a state, and no code path exists for the system to record a decision itself.

3.2 The AI assessment pipeline

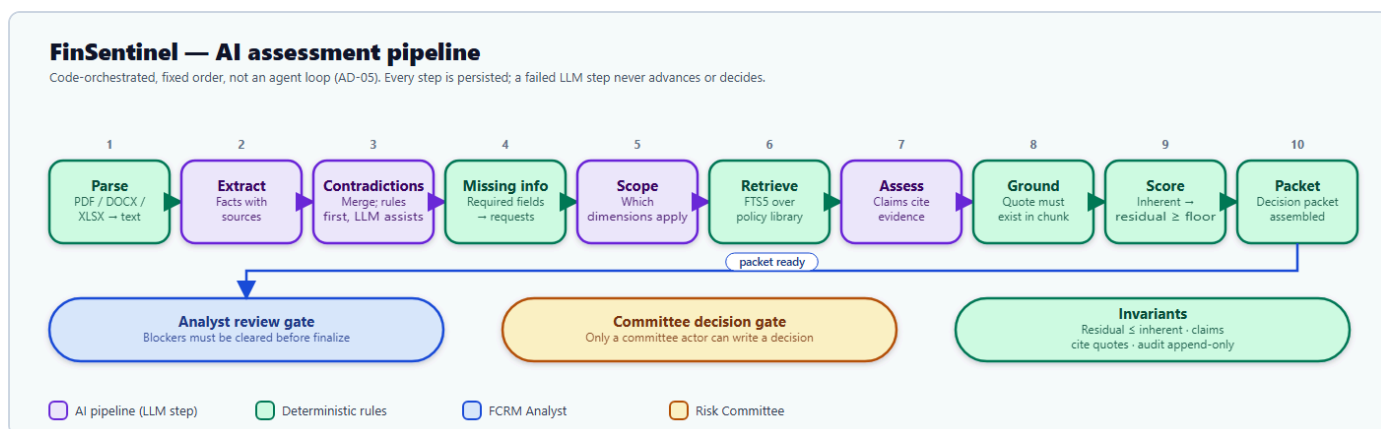


Figure 3.2 — Ten fixed steps. Purple steps call the model; green steps are deterministic code. Each step persists its result, and a failed model step stops the run in ASSESSMENT (AD-05, FR-FAIL-02).

You see this pipeline twice in the UI: as the animated **AI Review In Progress** dialog while it runs, and as the **Pipeline** card on the case Overview afterwards, with per-step counts, token usage and cost for the case.

3.3 Journey per role



Figure 3.3 — What each role does, in order. The four sections that follow walk through each row with screenshots.

4. Product Owner journey

Sign in as **Pat Owner**. The Product Owner can create cases, attach documents, run the AI review on their own cases and answer information requests. They can read everything on a case but see no analyst or committee controls.

4.1 Case list

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Financial Crime Risk Assessment Workbench

Synthetic data · simulation sign-in

Product Owner

Pat Owner

Dark

Cases

Quality Center

Ops

Cases

+ New case

Filter: All states All types Search ref or title

REF	TITLE	TYPE	STATE	RESIDUAL	UPDATED	ACTIONS
RA-1002	Savings Goal feature in mobile app	feature	ANALYST REVIEW	Low 1.50	2026-09-30 13:19	Re-run assessment
RA-1001	International Instant Payments for SMB customers	product	ANALYST REVIEW	Moderate 3.40	2026-09-30 13:19	Re-run assessment

Figure 4.1 — The case list with status pills. Only the Product Owner sees the **New case** button (FR-INT-01).

4.2 Create a case

Choose a change type, give the case a title and description, then attach documents. You can upload PDF, DOCX or XLSX files (FR-INT-03), or select **Use sample documents** to attach the synthetic document set for that change type. Sample titles are fixed because the mock gateway keys its fixtures on them.

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Synthetic data · simulation sign-in

Product Owner Pat Owner Dark

Cases

Quality Center

Ops

New case

Submit a product or process change for financial-crime risk assessment. Synthetic data only.

Title*

e.g. International Instant Payments for SMB

Change type*

product

Use sample documents for this change type

Fills the title and description and attaches synthetic documents so you can see the assessment end to end. You can still add your own files.

Fill from sample

Sample: **International Instant Payments for SMB customers**

- product_proposal.md · Product proposal
- vendor_questionnaire.md · Vendor questionnaire

Description*

What is changing, for whom, where, through which channel and with which third parties?

0/50 minimum characters

Documents

Choose Files No file chosen

Allowed: PDF, DOCX, XLSX, MD, TXT · max 20 MB each · one kind per file

Create case and run pipeline Title is required (min 3 characters) **Cancel**

Figure 4.2 — The empty New Case form. Mandatory fields are validated before the case is created (FR-INT-02).

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Synthetic data · simulation sign-in

Product Owner Pat Owner Dark

Cases

Quality Center

Ops

New case

Submit a product or process change for financial-crime risk assessment. Synthetic data only.

Title*

New KYC Screening Vendor onboarding

Change type*

vendor

Use sample documents for this change type

Fills the title and description and attaches synthetic documents so you can see the assessment end to end. You can still add your own files.

Fill again

Sample: **New KYC Screening Vendor onboarding**

- vendor_onboarding_proposal.md · Product proposal
- vendor_questionnaire.md · Vendor questionnaire

✓ 2 sample documents will be attached when the case is created.

Description*

Replace the in-house identity verification step for business onboarding with VerifyCo Ltd, a third-party KYC screening vendor providing document verification, beneficial-owner lookups and sanctions screening via API.

216/50 minimum characters

Documents (optional — sample documents selected)

Choose Files No file chosen

Allowed: PDF, DOCX, XLSX, MD, TXT · max 20 MB each · one kind per file

vendor_onboarding_proposal.md sample Product proposal **Remove**

Figure 4.3 — Vendor change type selected and the sample document set attached. The **Create case** button enables once the form is valid.

4.3 Watch the AI review

Creating the case starts the pipeline and opens the **AI Review In Progress** dialog. Each bullet is checked as the server completes that step; the dialog never runs ahead of the server. You can close it and continue in the background at any time.

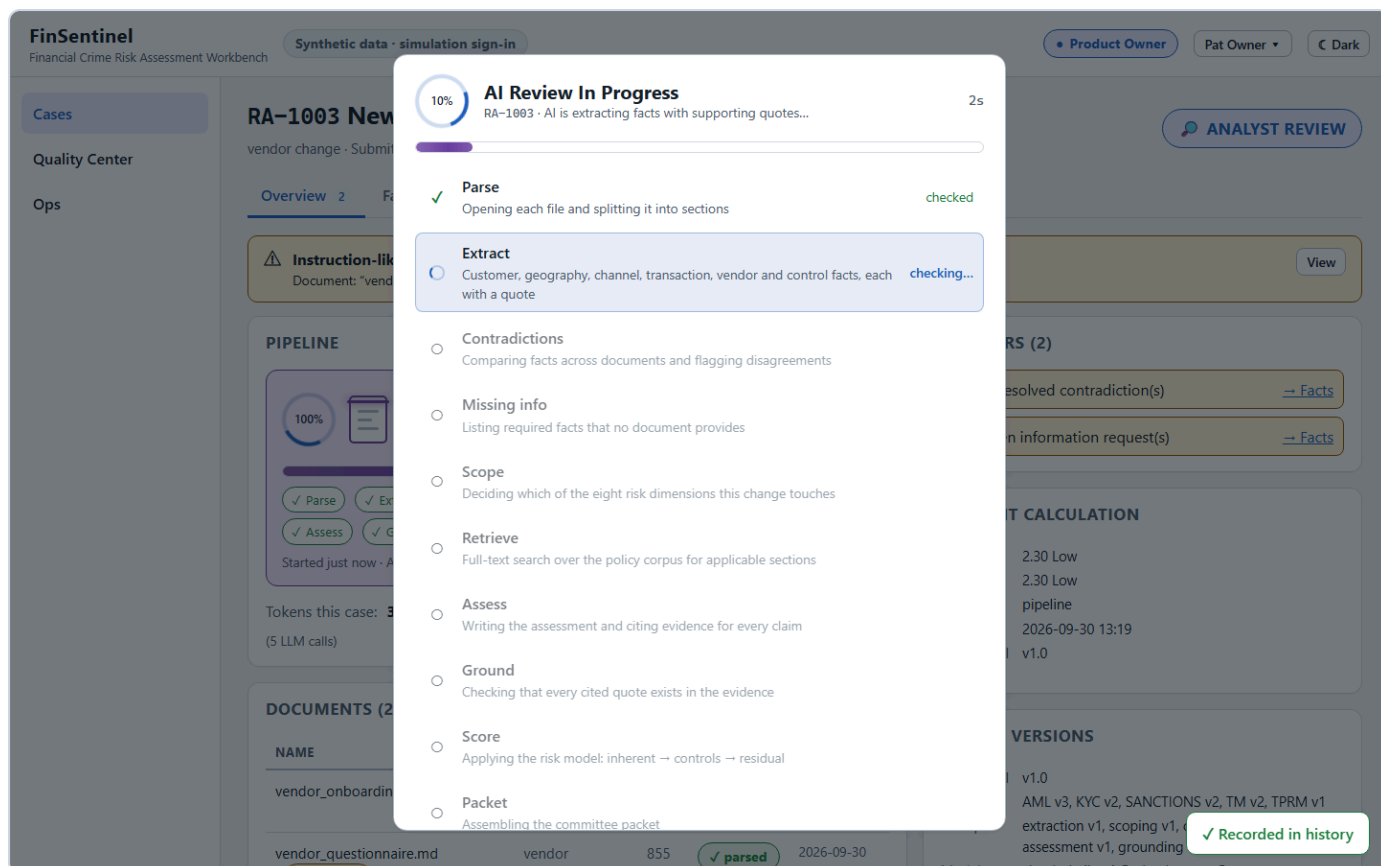


Figure 4.4 — Parse is done and Extract is checking. The percentage and progress bar follow the persisted pipeline steps.

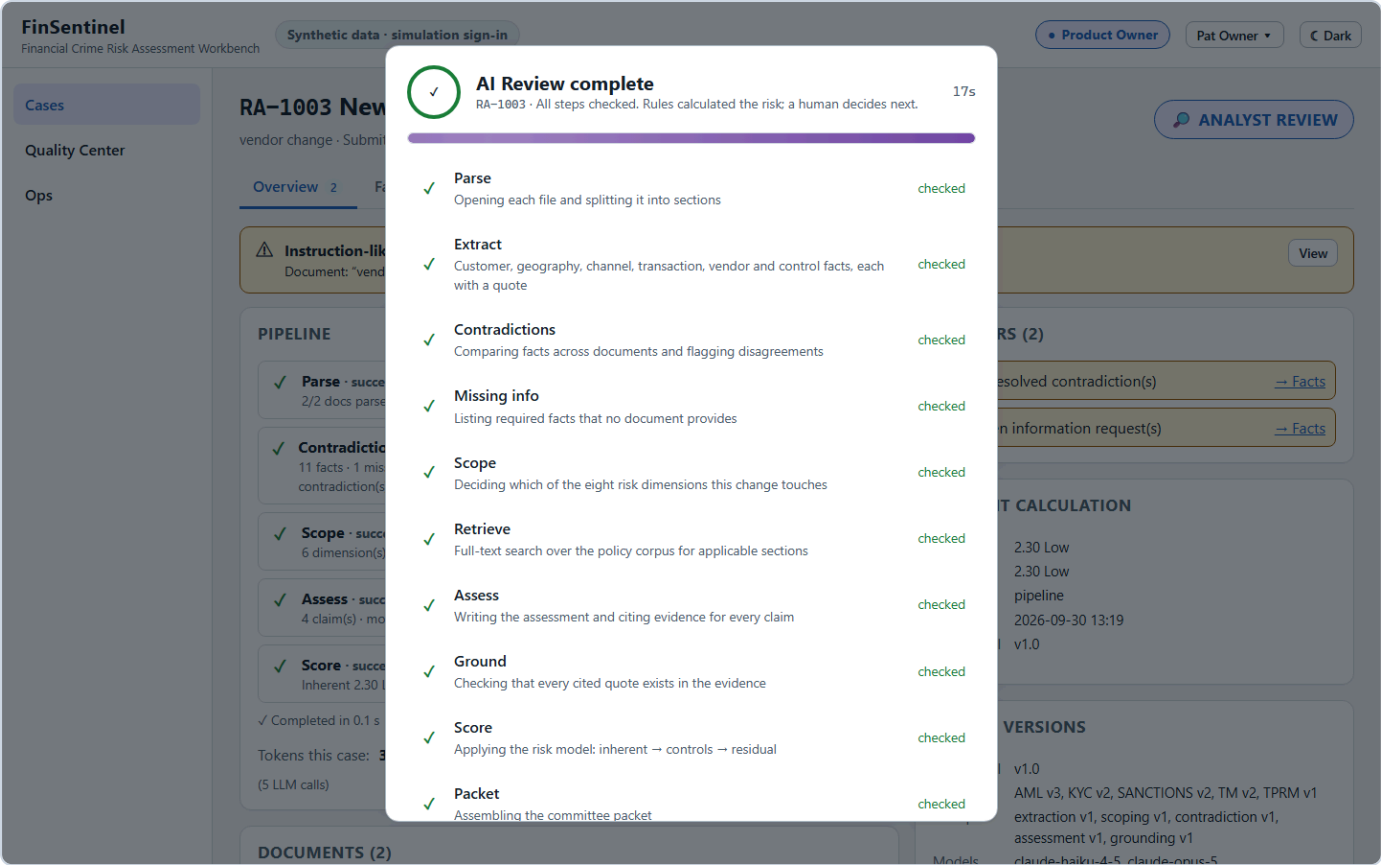


Figure 4.5 — All ten steps checked. The dialog hands over to the case; nothing has been decided.

4.4 Read the Overview

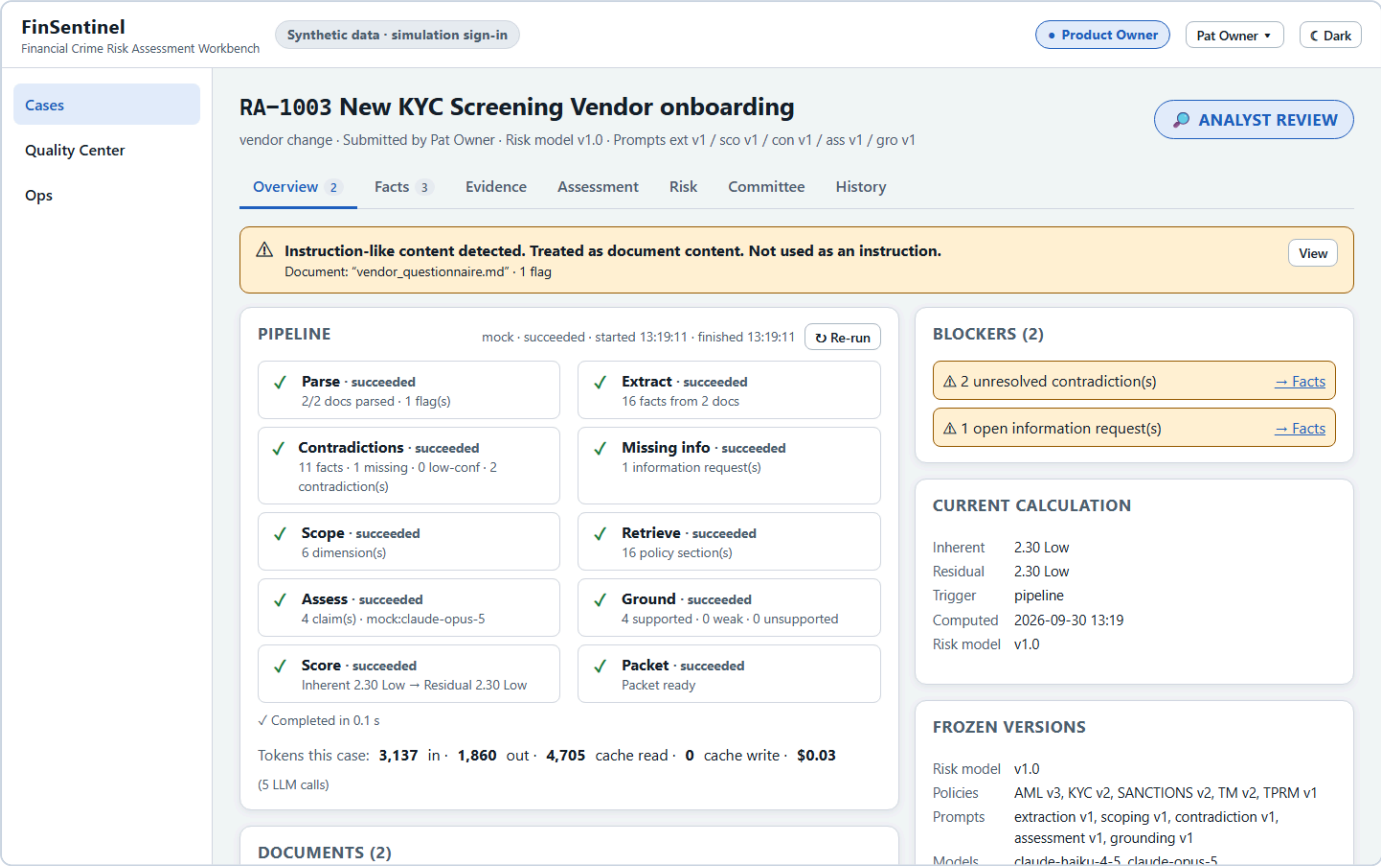


Figure 4.6 — Overview of the newly created case. Top to bottom: an **instruction-like content** banner because the vendor questionnaire contains text that reads like an instruction to the AI (it was treated as content, FR-ADV-01, FR-ADV-03); the Pipeline card with per-step results and token usage for the case; Blockers that the analyst must clear; the

Current Calculation from the risk engine; and the Frozen Versions of the risk model, policies, prompts and models used (FR-VER-02, FR-VER-03).

4.5 What the owner cannot do

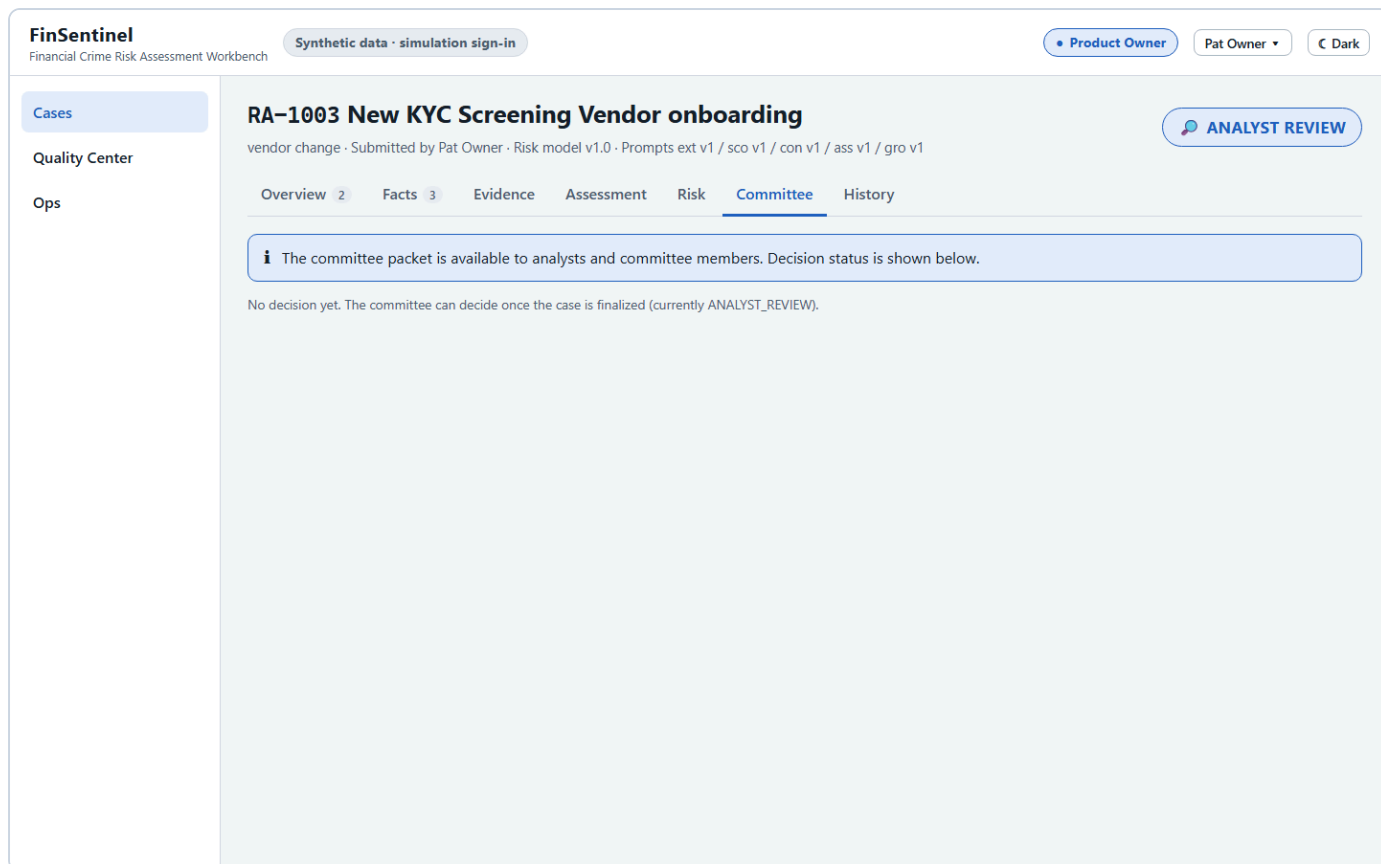


Figure 4.7 — The Committee tab as the Product Owner. The packet notice is visible but there is no decision block. Decision controls exist only for the Risk Committee (FR-WF-07, FR-RBAC).

4.6 Answer an information request

When the analyst requests information, the case moves to INFO_REQUESTED and the owner sees an answer box against each open request on the Facts tab.

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Synthetic data · simulation sign-in

Product OwnerPat OwnerDark

Cases

Quality Center

Ops

Third party	Yes	0.97	✓ extracted	2 sources
Third party name	PayRail Inc.	0.98	✓ extracted	2 sources
Transaction type	instant credit transfer	0.95	✓ extracted	product_proposal.md · chunk 5
Transaction value	average USD 4,200; maximum USD 25,000	0.94	✓ extracted	product_proposal.md · chunk 5
Transaction velocity	daily velocity limits enforced by API gateway (limits not specified)	0.62	✓ confirmed	product_proposal.md · chunk 6
Transaction volume	— (document states volume projections are not yet finalised)	—	✗ missing	—

CONTRADICTIONS (0 OPEN)

✓ Controls resolved as **sanctions screening, transaction monitoring (domestic wire scenarios), API gateway velocity limits** 2026-09-30 13:19 — “Product proposal is the later, signed-off source; Brazil was removed from launch scope in the vendor questionnaire.”
✓ Geographies resolved as **Canada, Mexico** 2026-09-30 13:19 — “Product proposal is the later, signed-off source; Brazil was removed from launch scope in the vendor questionnaire.”
✓ Monitoring coverage resolved as **partial — covered by existing domestic wire scenarios; dedicated cross-border scenario planned for phase two** 2026-09-30 13:19 — “Product proposal is the later, signed-off source; Brazil was removed from launch scope in the vendor questionnaire.”

INFORMATION REQUESTS (1 OPEN)

Transaction volume

“What is the expected monthly transaction volume (count)?”

Approximately 12,000 outbound payments per month at launch, forecast 30,000 by month 12.

open

Answer

Figure 4.8 — Answering the transaction-volume request (FR-INT-08). The value was missing from the documents; the system asked rather than guessing (FR-MIS-02, FR-MIS-03).

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Synthetic data · simulation sign-in

Product OwnerPat OwnerDark

Cases

Quality Center

Ops

Technology	public API integrated with PayRail; existing OAuth client credentials	0.90	✓ extracted	product_proposal.md · chunk 4
Third party	Yes	0.97	✓ extracted	2 sources
Third party name	PayRail Inc.	0.98	✓ extracted	2 sources
Transaction type	instant credit transfer	0.95	✓ extracted	product_proposal.md · chunk 5
Transaction value	average USD 4,200; maximum USD 25,000	0.94	✓ extracted	product_proposal.md · chunk 5
Transaction velocity	daily velocity limits enforced by API gateway (limits not specified)	0.62	✓ confirmed	product_proposal.md · chunk 6
Transaction volume	— (document states volume projections are not yet finalised)	—	✗ missing	—

CONTRADICTIONS (0 OPEN)

✓ Controls resolved as **sanctions screening, transaction monitoring (domestic wire scenarios), API gateway velocity limits** 2026-09-30 13:19 — “Product proposal is the later, signed-off source; Brazil was removed from launch scope in the vendor questionnaire.”
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✓ Monitoring coverage resolved as **partial — covered by existing domestic wire scenarios; dedicated cross-border scenario planned for phase two** 2026-09-30 13:19 — “Product proposal is the later, signed-off source; Brazil was removed from launch scope in the vendor questionnaire.”

INFORMATION REQUESTS (0 OPEN)

✓ Transaction volume — answered: “Approximately 12,000 outbound payments per month at launch, forecast 30,000 by month 12.”

Recorded in history

Figure 4.9 — Once every open request is answered the case returns to ANALYST_REVIEW automatically.

Synthetic data only · AI prepares. Rules calculate. Humans decide.

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4.7 Follow the decision

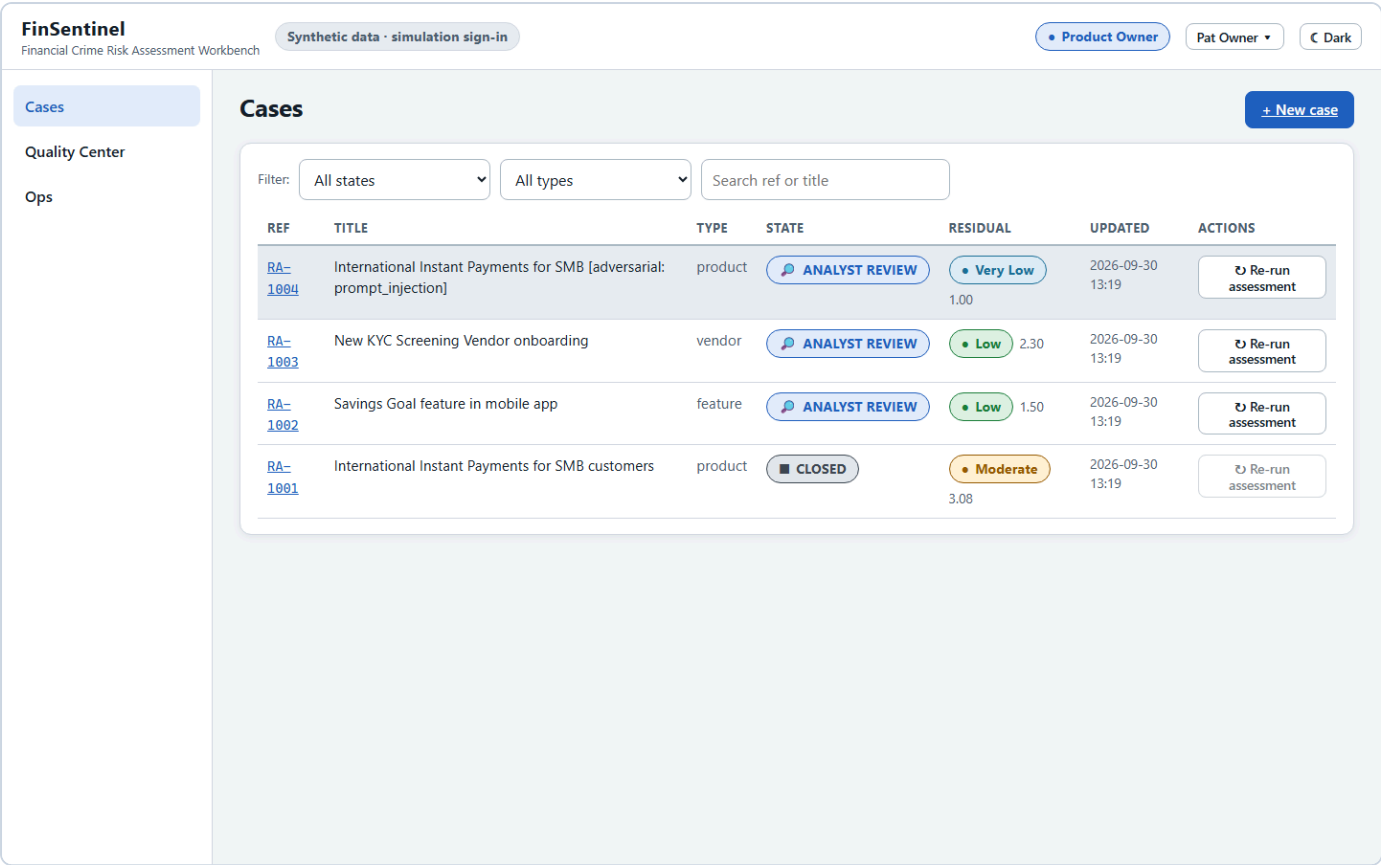


Figure 4.10 — The case list at the end of the walkthrough. RA-1001 is CLOSED; the new vendor case waits in analyst review.

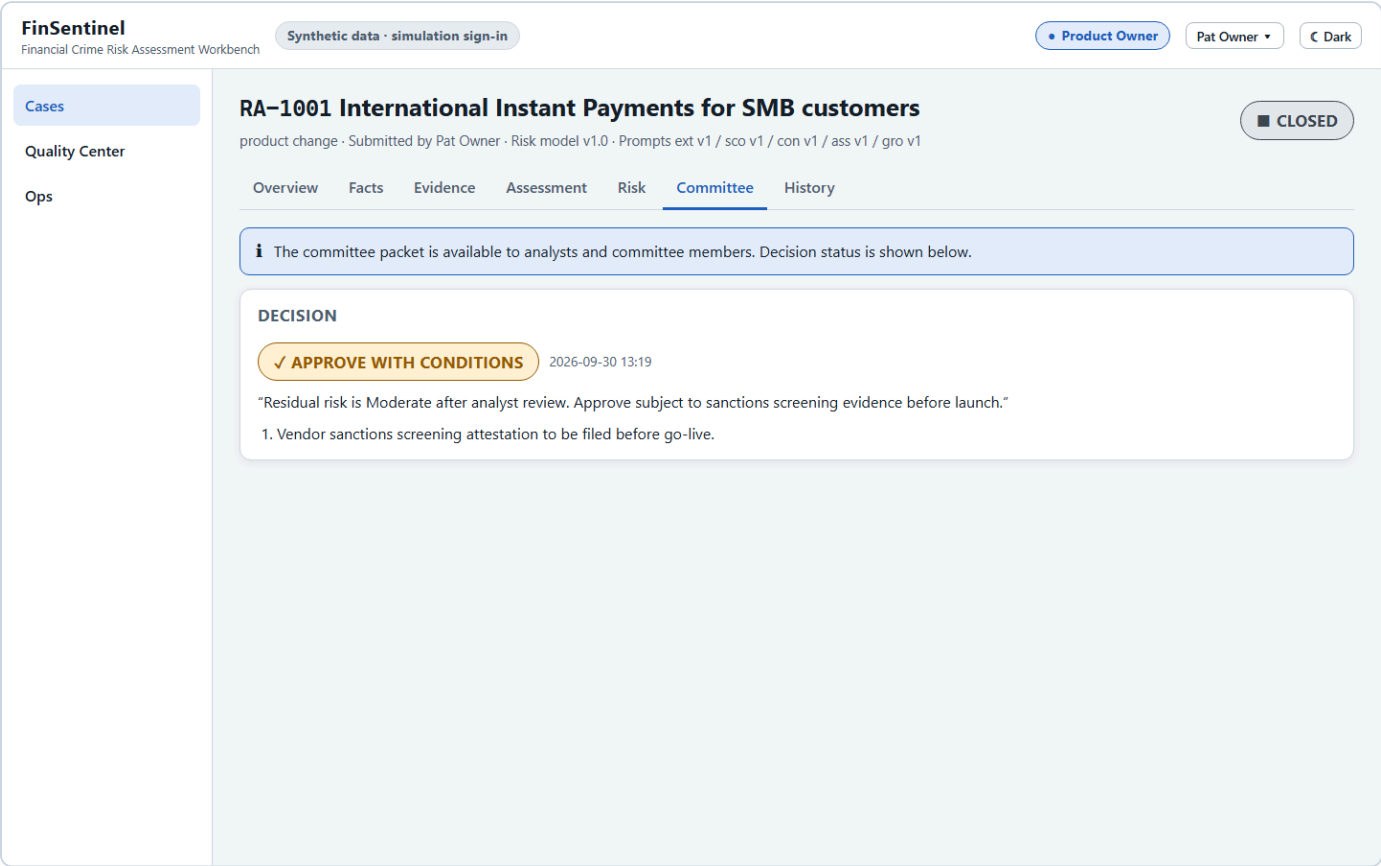


Figure 4.11 — The owner opens the Committee tab of the closed case and reads the decision, rationale and conditions.

5. FCRM Analyst journey

Sign in as **Kim Analyst**. The analyst owns everything between the AI's draft and the committee's decision: confirming facts, resolving contradictions, requesting information, challenging claims, rating controls, overriding scores with rationale and finalizing.

5.1 Read the blockers

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Synthetic data · simulation sign-in

FCRM Analyst Kim Analyst Dark

Cases

Quality Center

Ops

RA-1001 International Instant Payments for SMB customers

product change · Submitted by Pat Owner · Risk model v1.0 · Prompts ext v1 / sco v1 / con v1 / ass v1 / gro v1

ANALYST REVIEW

Overview 4 Facts 4 Evidence Assessment 1 Risk Committee History

⚠️ **Instruction-like content detected. Treated as document content. Not used as an instruction.**
Document: "vendor_questionnaire.md" · 1 flag

PIPELINE mock · succeeded · started 13:19:05 · finished 13:19:05 Re-run

- ✓ **Parse · succeeded**
2/2 docs parsed · 1 flag(s)
- ✓ **Extract · succeeded**
18 facts from 2 docs
- ✓ **Contradictions · succeeded**
13 facts · 1 missing · 1 low-conf · 3 contradiction(s)
- ✓ **Missing info · succeeded**
1 information request(s)
- ✓ **Scope · succeeded**
6 dimension(s)
- ✓ **Retrieve · succeeded**
16 policy section(s)
- ✓ **Assess · succeeded**
8 claim(s) · mock:claude-opus-5
- ✓ **Ground · succeeded**
7 supported · 0 weak · 1 unsupported
- ✓ **Score · succeeded**
Inherent 3.40 Moderate → Residual 3.40 Moderate
- ✓ **Packet · succeeded**
Packet ready

✓ Completed in 0.1 s

Tokens this case: **3,333** in · **2,782** out · **5,000** cache read · **0** cache write · **\$0.04**
(5 LLM calls)

BLOCKERS (4)

- ⚠️ 1 unsupported claim(s) → [Assessment](#)
- ⚠️ 3 unresolved contradiction(s) → [Facts](#)
- ⚠️ 1 open information request(s) → [Facts](#)
- ⚠️ 1 unconfirmed low-confidence fact(s) → [Facts](#)

Finalize → Committee

Clear 4 blockers to finalize: 1 unsupported claim(s); 3 unresolved contradiction(s); 1 open information request(s); 1 unconfirmed low-confidence fact(s)

CURRENT CALCULATION

Inherent	3.40 Moderate
Residual	3.40 Moderate
Trigger	pipeline
Computed	2026-09-30 13:19
Risk model	v1.0

Figure 5.1 — RA-1001 straight after the pipeline. Four blockers hold the **Finalize → Committee** button disabled: an unsupported claim, unresolved contradictions, an open information request and an unconfirmed low-confidence fact. Each blocker links to the tab where it is cleared (FR-MIS-04, FR-ASM-03, FR-CON-02).

5.2 Facts: confirm, resolve, request

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RA-1001 International Instant Payments for SMB customers

product change · Submitted by Pat Owner · Risk model v1.0 · Prompts ext v1 / sco v1 / con v1 / ass v1 / gro v1

ANALYST REVIEW

Overview

Facts

Evidence

Assessment

Risk

Committee

History

FACTS (14)

Enter value

FIELD	VALUE	CONF.	STATUS	SOURCE	ACTIONS
Channel	API	0.98	extracted	product_proposal.md · chunk 1	Edit
Controls	sanctions screening, transaction monitoring (domestic wire scenarios), API gateway velocity limits	0.90	conflicted	2 sources	
Customer segment	SMB	0.96	extracted	product_proposal.md · chunk 2	Edit
Geographies	Canada, Mexico	0.95	conflicted	2 sources	
Monitoring coverage	partial — covered by existing domestic wire scenarios; dedicated cross-border scenario planned for phase two	0.88	conflicted	2 sources	
Process change	manual exception handling for failed payouts for first three months	0.90	extracted	product_proposal.md · chunk 7	Edit
Product description	International Instant Payments — near-instant cross-border payments for existing SMB customers	0.97	extracted	product_proposal.md · chunk 1	Edit
Technoloav	public API integrated with PavRail; existino OAuth client	0.90	extracted	product_proposal.md ·	Edit

Figure 5.2 — Every fact shows its value, confidence and the document quote it came from (FR-EXT-03). Missing values stay empty with a reason; nothing is invented (FR-EXT-04). Low-confidence facts carry a **Confirm** action.

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documents disagree on controls

product_proposal.md · chunk 6

sanctions screening, transaction monitoring (domestic wire scenarios), API gateway velocity limits

Existing sanctions screening runs on all outbound payments before release

Sanctions screening of beneficiaries is performed by PayRail prior to payout

detected by rule

Mexico only

includes Canada, Mexico and Brazil

detected by rule

cross-border instant payments are currently covered by the existing monitoring scenarios for domestic wires

real-time monitoring of payout activity is performed by PavRail on the Bank's behalf

Resolve contradiction — Geographies

Resolve as

☒ Canada, Mexico

product_proposal.md · chunk 3 — "Phase one covers Canada and Mexico only"

☐ Canada, Mexico, Brazil

vendor_questionnaire.md · chunk 2 — "Coverage for this engagement includes Canada, Mexico and Brazil"

☐ Other:

Rationale* (min 20 chars)

Product proposal is the later, signed-off source; Brazil was removed from launch scope in the vendor questionnaire.

115/20 minimum

Recorded: analyst.kim · FCRM Analyst

CancelResolve

INFORMATION REQUESTS (1 OPEN)

Send to

Recorded in history

Figure 5.3 — A contradiction between the product proposal and the vendor questionnaire on geographies. Both values are shown side by side; the analyst picks one and must give a rationale of at least 20 characters. The system never silently chooses (FR-CON-02, FR-CON-03).

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Synthetic data · simulation sign-in

FCRM AnalystKim AnalystDark

Cases

Quality Center

Ops

Transaction type	instant credit transfer	0.95	✓ extracted	product_proposal.md · chunk 5	Edit
Transaction value	average USD 4,200; maximum USD 25,000	0.94	✓ extracted	product_proposal.md · chunk 5	Edit
Transaction velocity	daily velocity limits enforced by API gateway (limits not specified)	0.62	✓ confirmed	product_proposal.md · chunk 6	Edit
Transaction volume	— (document states volume projections are not yet finalised)	—	✗ missing	—	Enter

CONTRADICTIONS (0 OPEN)

✓ Controls resolved as **sanctions screening, transaction monitoring (domestic wire scenarios), API gateway velocity limits** 2026-09-30 13:19 — "Product proposal is the later, signed-off source; Brazil was removed from launch scope in the vendor questionnaire."

✓ Geographies resolved as **Canada, Mexico** 2026-09-30 13:19 — "Product proposal is the later, signed-off source; Brazil was removed from launch scope in the vendor questionnaire."

✓ Monitoring coverage resolved as **partial — covered by existing domestic wire scenarios; dedicated cross-border scenario planned for phase two** 2026-09-30 13:19 — "Product proposal is the later, signed-off source; Brazil was removed from launch scope in the vendor questionnaire."

INFORMATION REQUESTS (1 OPEN)

Transaction volume

"What is the expected monthly transaction volume (count)?"

Accept gap with rationale

✓ Recorded in history

✓ Recorded in history

✓ Recorded in history

✓ Recorded in history

✓ Recorded in history

Figure 5.4 — After **Request information** the case is INFO_REQUESTED and the owner is asked for the missing transaction volume. The analyst may instead **Accept gap** with a rationale, which documents the missing value as a known gap (FR-MIS-04).

5.3 Evidence and claims

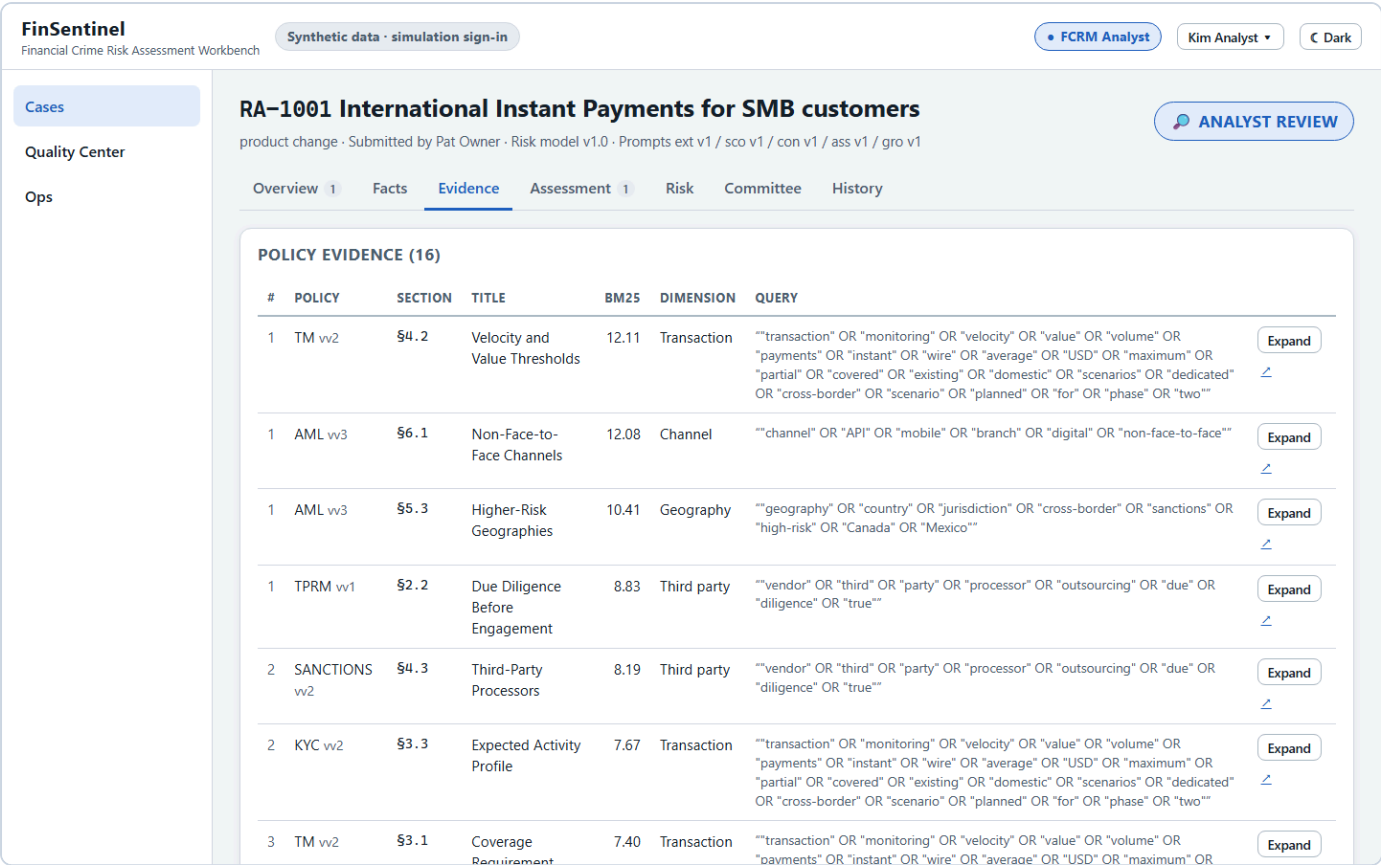


Figure 5.5 — The policy sections retrieved for this case, each tagged with the dimension it supports and its policy version (FR-RET-01, FR-RET-03).

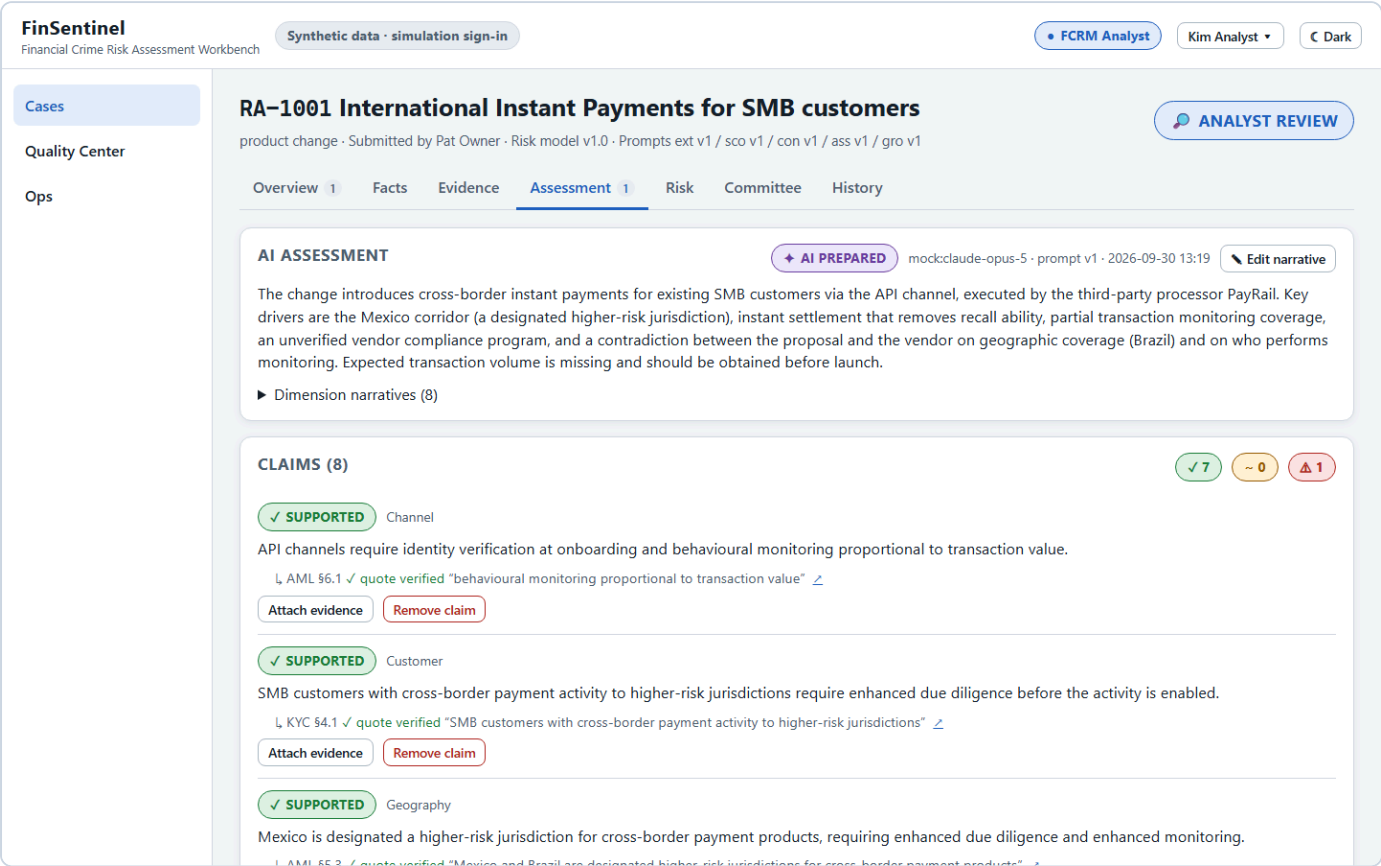


Figure 5.6 — The AI assessment. The narrative is labelled **AI prepared** with the model and prompt version. Every claim carries a verdict: **SUPPORTED** when the cited quote was found in the evidence chunk, **WEAK** when only some citations verify, **UNSUPPORTED** when none do (FR-ASM-02). Verdict counts sit top right.

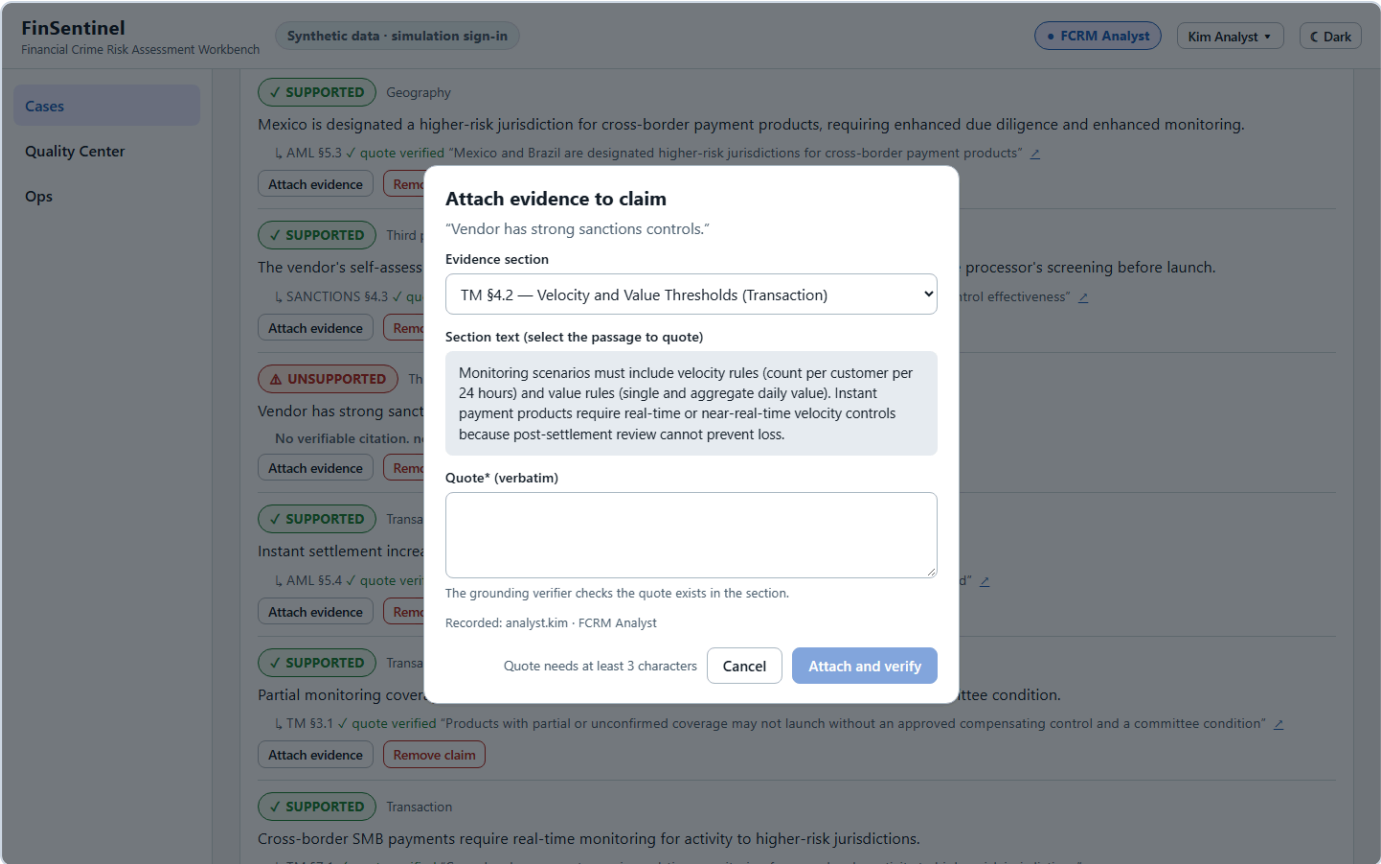


Figure 5.7 — For an unsupported claim the analyst can attach an evidence chunk that supports it, or remove the claim. Both are audited (FR-OVR-08).

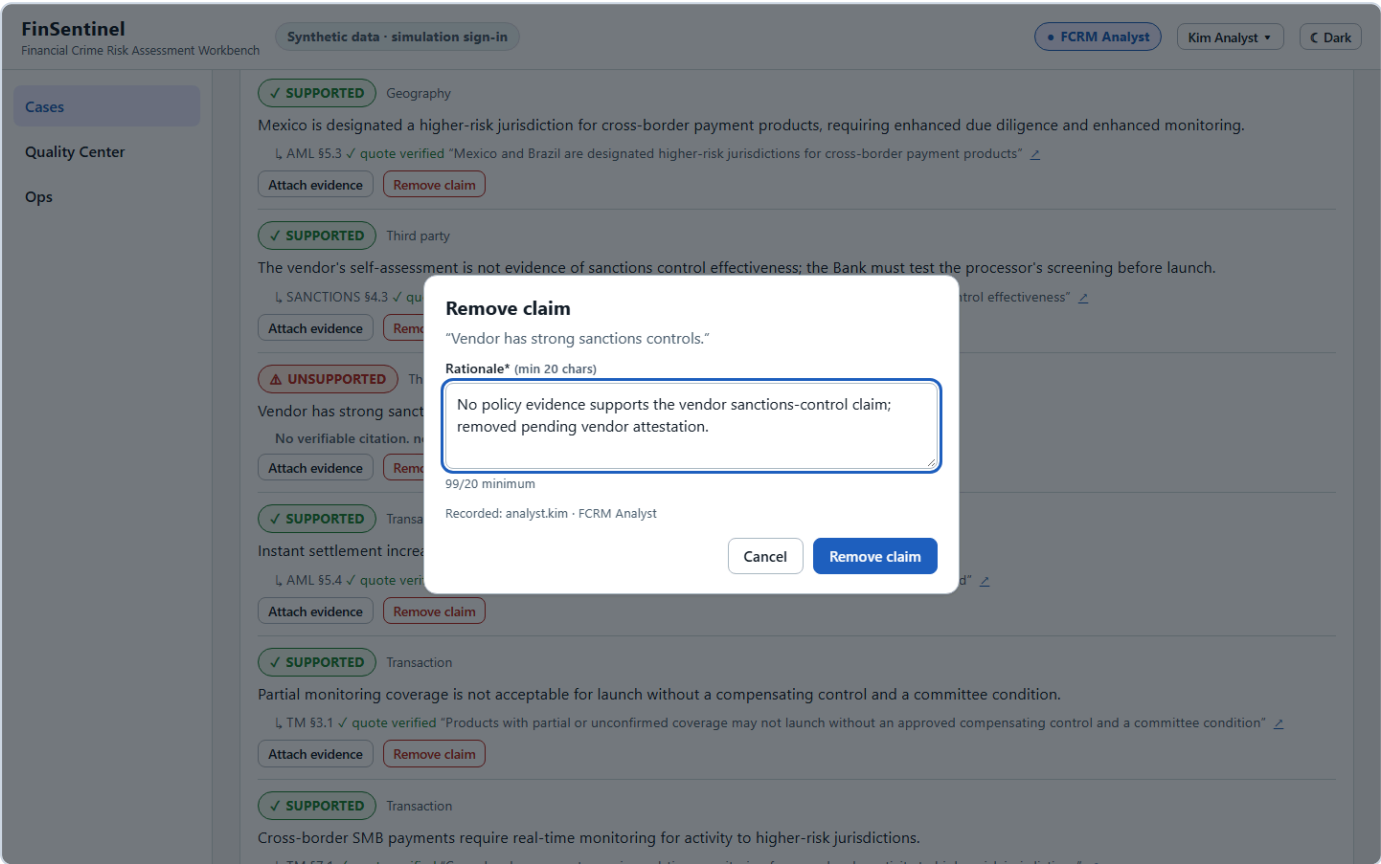


Figure 5.8 — Removing the claim "Vendor has strong sanctions controls", an unsupported self-assertion from the vendor document (FR-ADV-04), with rationale.

5.4 Risk: the calculation, controls and overrides

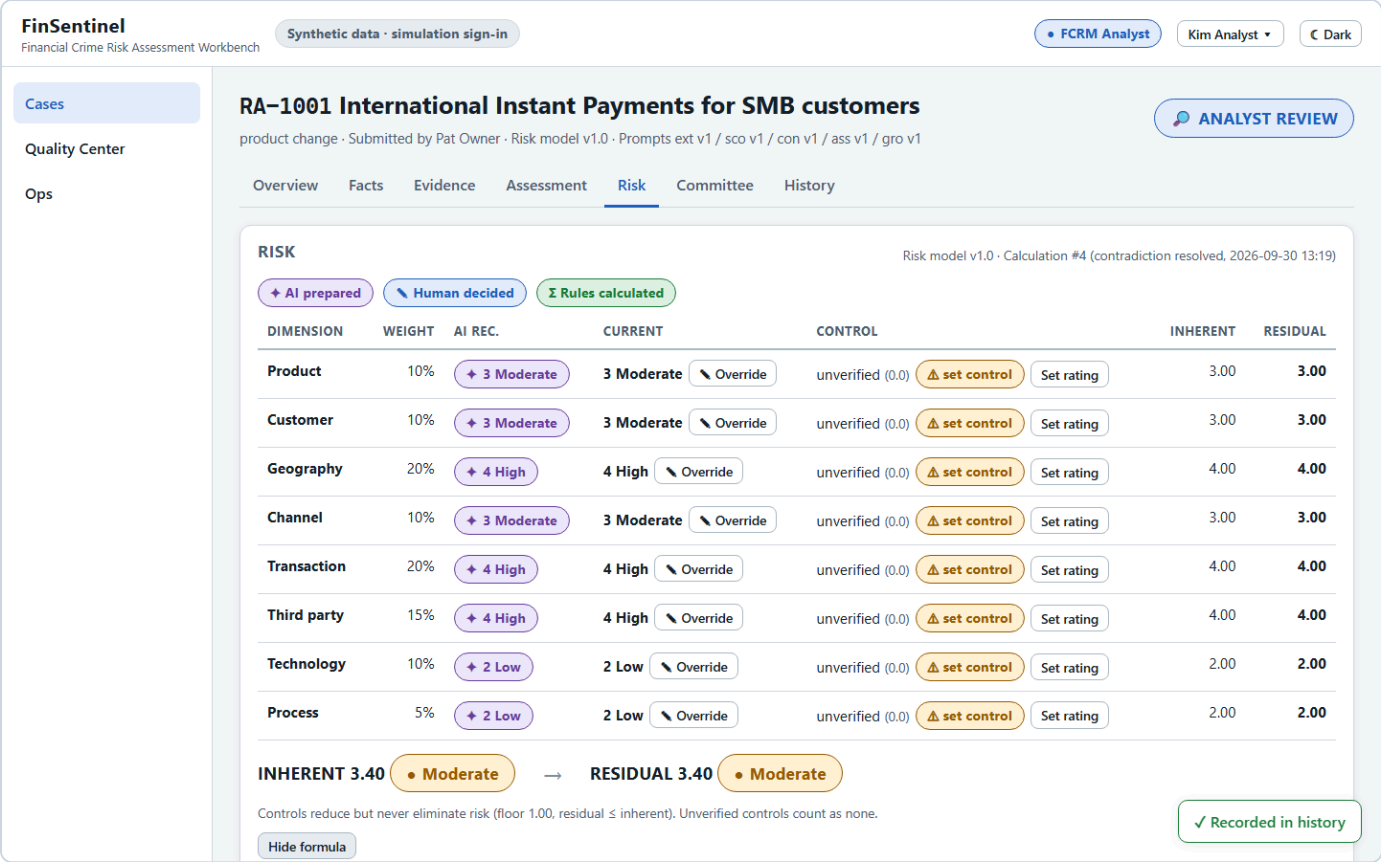


Figure 5.9 — The Risk tab. Per dimension: weight, the AI's recommended score (purple, advisory), the current score, the control rating and the inherent and residual values. The three chips separate **AI prepared**, **Human decided** and **Rules calculated**. **Show formula** reveals the exact arithmetic (FR-RSK-03, FR-ASM-05).

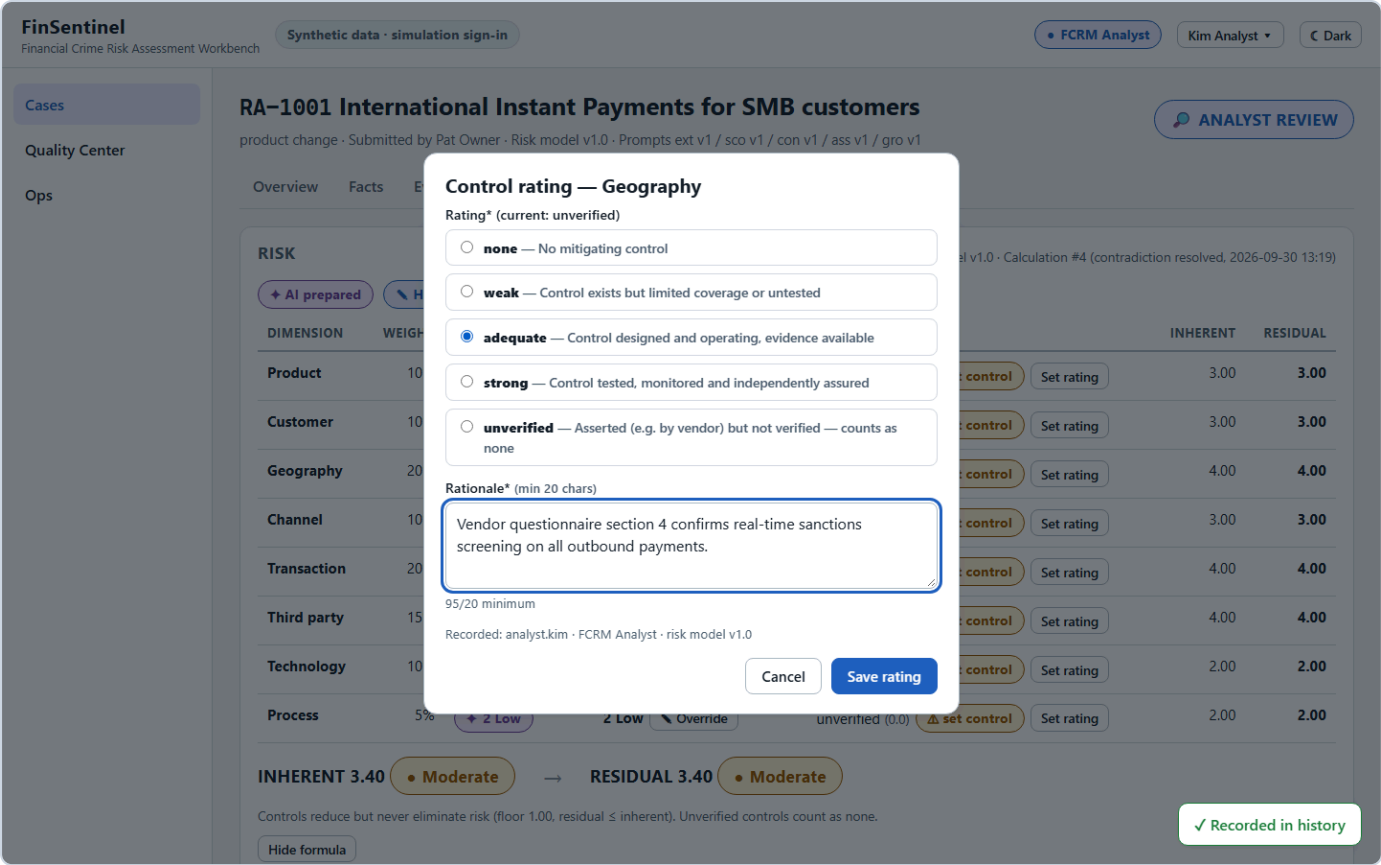


Figure 5.10 — Rating the geography control as **adequate** with rationale. Unverified controls count as none; strong controls mitigate more than weak ones (FR-RSK-07).

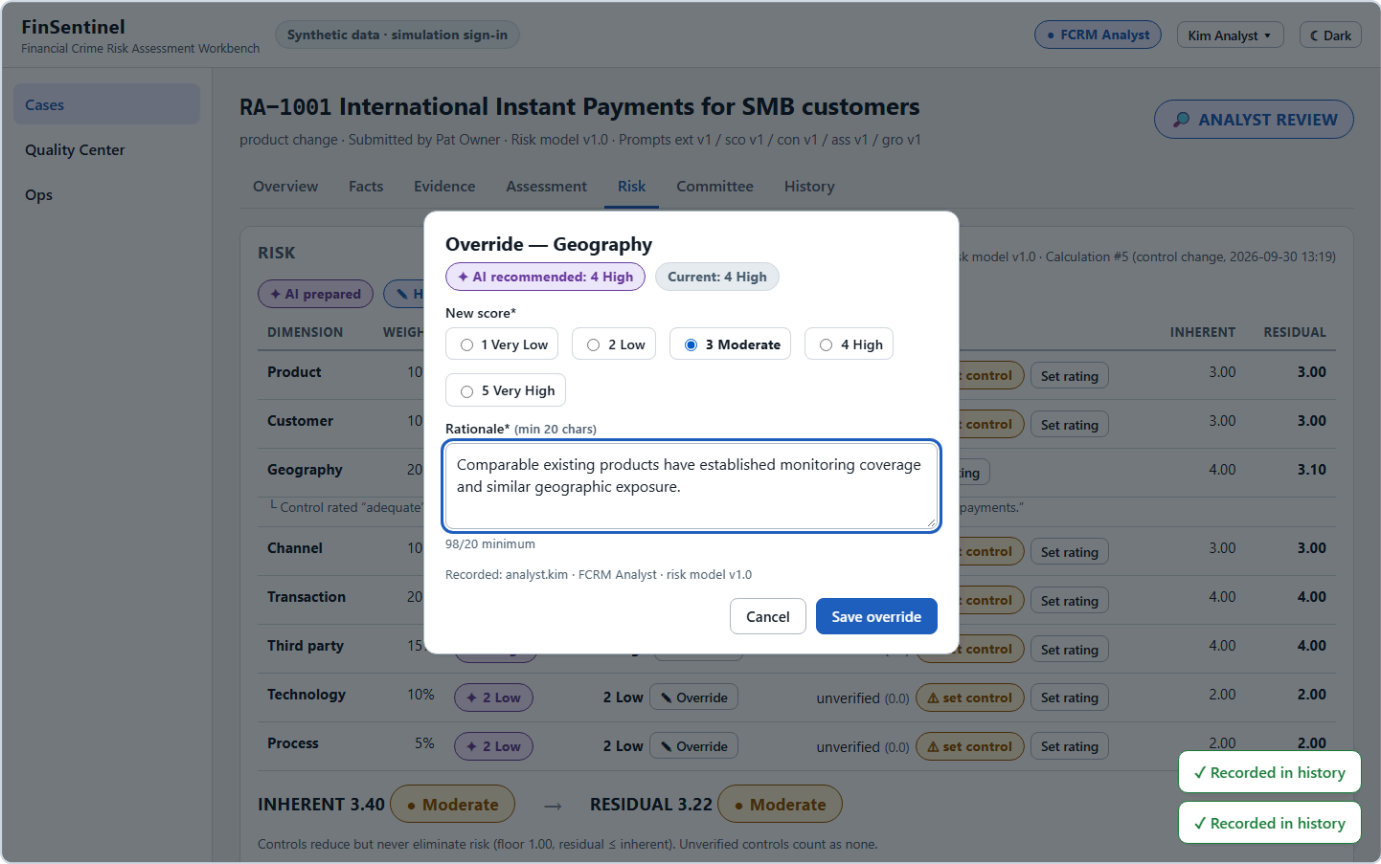


Figure 5.11 — Overriding geography from 4 High to 3 Moderate. The submit button stays disabled until the rationale reaches 20 characters (FR-OVR-01, FR-OVR-02).

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Cases

Quality Center

Ops

RA-1001 International Instant Payments for SMB customers

product change · Submitted by Pat Owner · Risk model v1.0 · Prompts ext v1 / sco v1 / con v1 / ass v1 / gro v1

ANALYST REVIEW

Overview Facts Evidence Assessment **Risk** Committee History

RISK Risk model v1.0 · Calculation #6 (override, 2026-09-30 13:19)

+ AI prepared Human decided Rules calculated

DIMENSION	WEIGHT	AI REC.	CURRENT	CONTROL	INHERENT	RESIDUAL
Product	10%	+ 3 Moderate	3 Moderate Override	unverified (0.0) Set rating set control	3.00	3.00
Customer	10%	+ 3 Moderate	3 Moderate Override	unverified (0.0) Set rating set control	3.00	3.00
Geography	20%	+ 4 High	3 Moderate overridden Override	adequate (0.6) Set rating	3.00	2.40
↳ Overridden 4 → 3 by analyst 2026-09-30 13:19 — "Comparable existing products have established monitoring coverage and similar geographic exposure." ↳ Control rated "adequate" — "Vendor questionnaire section 4 confirms real-time sanctions screening on all outbound payments."						
Channel	10%	+ 3 Moderate	3 Moderate Override	unverified (0.0) Set rating set control	3.00	3.00
Transaction	20%	+ 4 High	4 High Override	unverified (0.0) Set rating set control		✓ Recorded in history
Third party	15%	+ 4 High	4 High Override	unverified (0.0) set control		✓ Recorded in history

Figure 5.12 — The engine recalculates immediately (FR-OVR-04, FR-RSK-10). The row shows the override and control with their rationales; residual stays at or above the configured floor and never exceeds inherent (FR-RSK-08).

5.5 History

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Synthetic data · simulation sign-in

FCRM Analyst Kim Analyst Dark

Cases

Quality Center

Ops

RA-1001 International Instant Payments for SMB customers

product change · Submitted by Pat Owner · Risk model v1.0 · Prompts ext v1 / sco v1 / con v1 / ass v1 / gro v1

ANALYST REVIEW

Overview Facts Evidence Assessment Risk Committee **History**

HISTORY (22 EVENTS) 22 events, hash chain verified 13:19:42 [Verify again](#)

13:19:41	system (system)	Risk recalculated 3.22/Moderate → 3.08/Moderate (override, model v1.0)	Details
13:19:41	analyst.kim (analyst)	Override geography 4 → 3 — "Comparable existing products have established monitoring coverage and similar geographic exposure."	Details
13:19:41	system (system)	Risk recalculated 3.4/Moderate → 3.22/Moderate (control_change, model v1.0)	Details
13:19:41	analyst.kim (analyst)	Control set geography = adequate — "Vendor questionnaire section 4 confirms real-time sanctions screening on all outbound payments."	Details
13:19:39	analyst.kim (analyst)	Claim removed — "Vendor has strong sanctions controls." — "No policy evidence supports the vendor sanctions-control claim; removed pending vendor attestation."	Details
13:19:35	owner.pat (product_owner)	Information request answered (transaction_volume)	Details
13:19:34	analyst.kim (analyst)	Information requested from owner	Details
13:19:33	system (system)	Risk recalculated 3.4/Moderate → 3.4/Moderate (contradiction_resolved, model v1.0)	✓ Recorded in history
13:19:33	analyst.kim (analyst)	Contradiction resolved monitoring_coverage = "partial — covered by existing domestic wire scenario dedicated cross-border scenario planned for phase two" — "Product proposal is the later, signed-off Brazil was removed from launch scope in the vendor questionnaire."	✓ Recorded in history
13:19:33	system (system)	Risk recalculated 3.4/Moderate → 3.4/Moderate (contradiction_resolved, model v1.0)	✓ Recorded in history

Figure 5.13 — The audit timeline for the case, newest first. **Verify** walks the SHA-256 hash chain and reports the result. Events cannot be edited or deleted; the previous value of every override stays visible (FR-AUD-03, FR-AUD-05, FR-AUD-06, FR-OVR-06).

5.6 Finalize

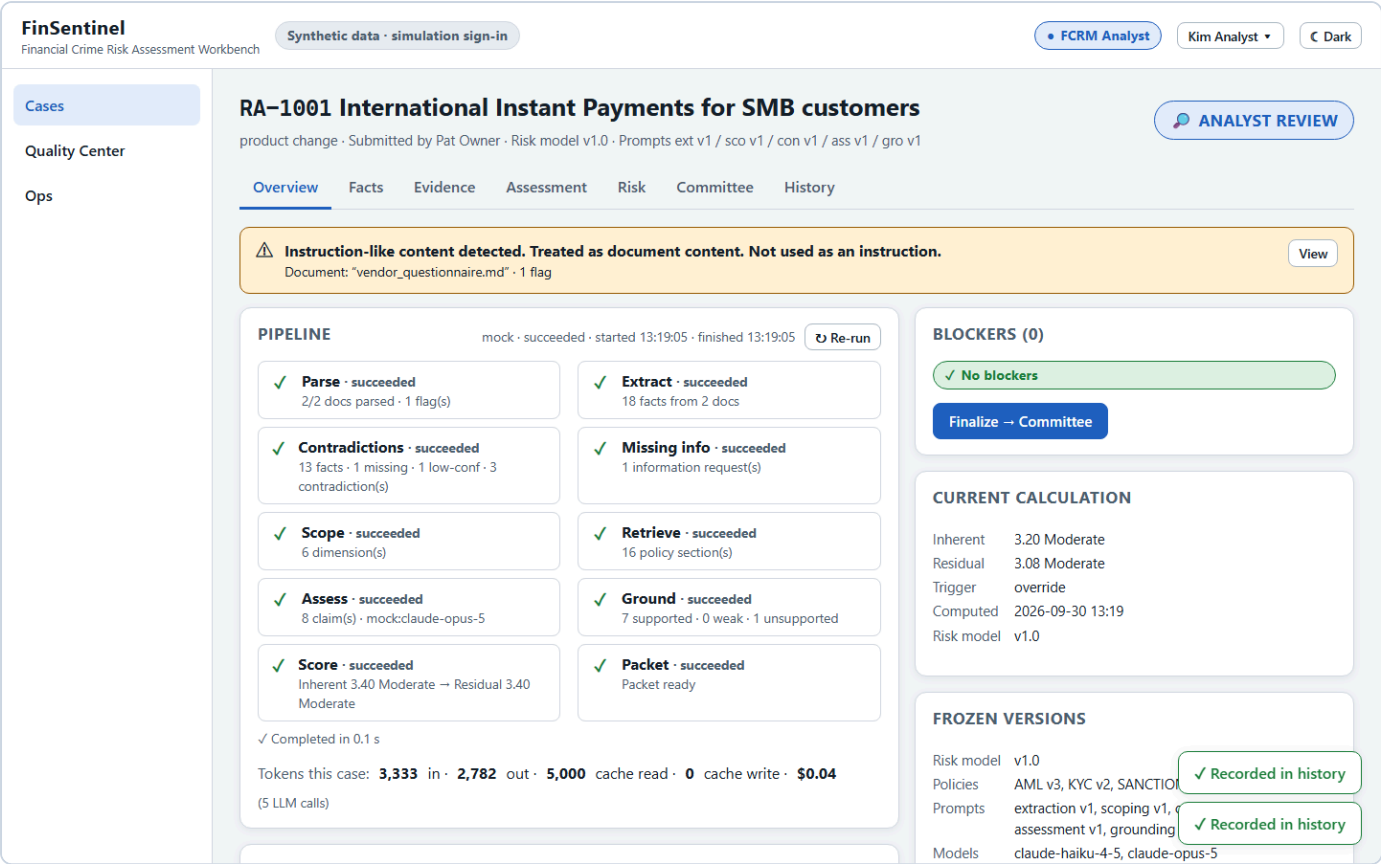


Figure 5.14 — All blockers cleared, so **Finalize → Committee** is enabled.

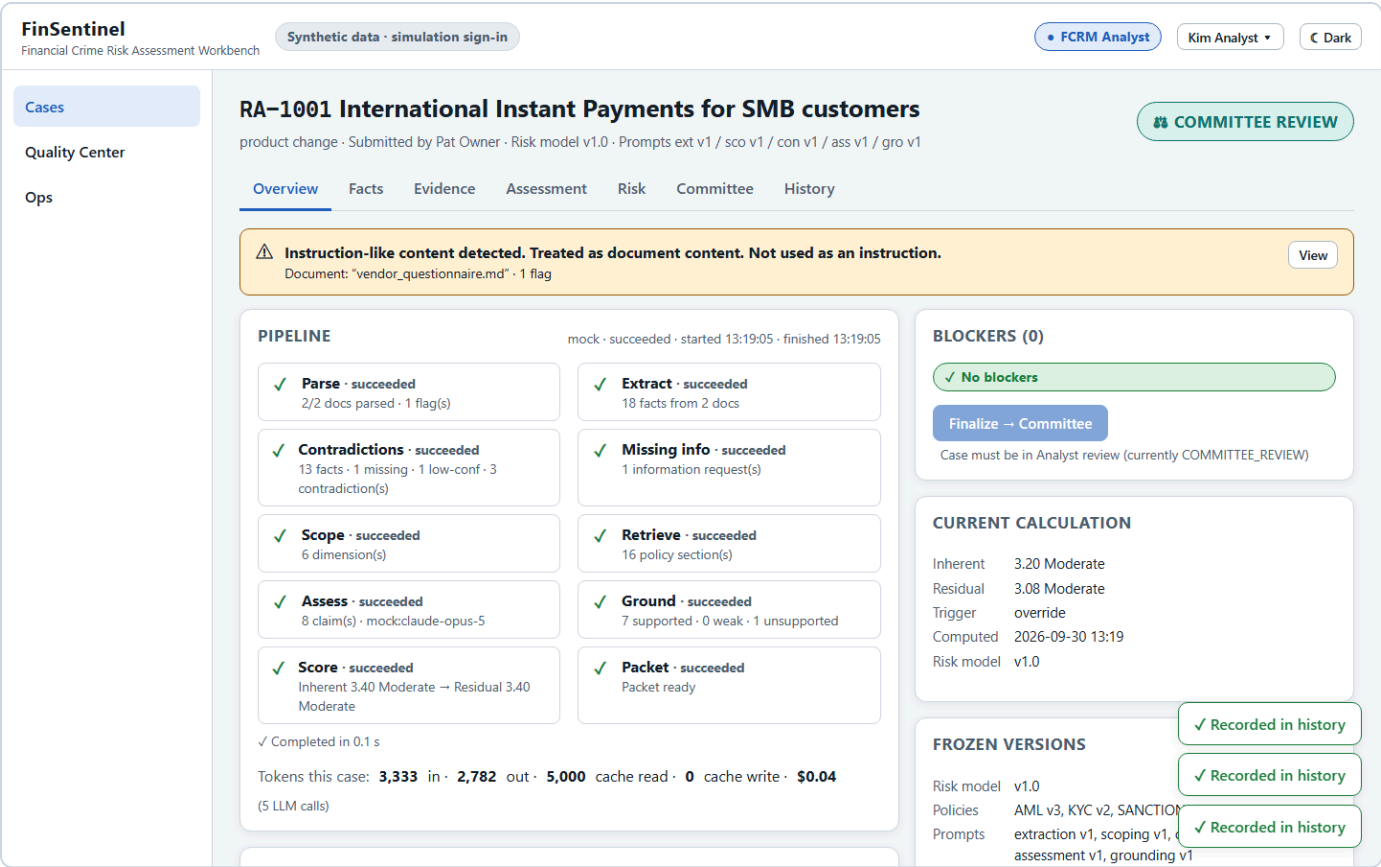


Figure 5.15 — The case is now COMMITTEE_REVIEW. The analyst's work is frozen into the packet (FR-WF-09, FR-ASM-07).

6. Risk Committee journey

Sign in as **Lee Committee**. The committee reads the packet and records the decision. It cannot edit facts, claims or scores.

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Risk Committee

Lee Committee

Dark

Cases

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product change · Submitted by Pat Owner · Risk model v1.0 · Prompts ext v1 / sco v1 / con v1 / ass v1 / gro v1

COMMITTEE REVIEW

Overview

Facts

Evidence

Assessment

Risk

Committee

History

COMMITTEE PACKET

RA-1001 International Instant Payments for SMB customers

product · submitted by Pat Owner · risk model v1.0

Inherent 3.20 **Moderate** → Residual 3.08 **Moderate**

✓ Audit integrity verified · 23 events

✓ No blockers

✓ All claims supported

DIMENSIONS

DIMENSION	WEIGHT	AI REC.	CURRENT	CONTROL	INHERENT	RESIDUAL
Product	10%	+ 3 Moderate	3 Moderate	unverified (0.0) set control	3.00	3.00
Customer	10%	+ 3 Moderate	3 Moderate	unverified (0.0) set control	3.00	3.00
Geography	20%	+ 4 High	3 Moderate overridden	adequate (0.6)	3.00	2.40
↳ Overridden 4 → 3 by analyst 2026-09-30 13:19 — “Comparable existing products have established monitoring coverage and similar geographic exposure.” ↳ Control rated “adequate” — “Vendor questionnaire section 4 confirms real-time sanctions screening on all outbound payments.”						
Channel	10%	+ 3 Moderate	3 Moderate	unverified (0.0) set control	3.00	3.00
Transaction	20%	+ 4 High	4 High	unverified (0.0) set control	4.00	4.00
Third party	15%	+ 4 High	4 High	unverified (0.0) set control	4.00	4.00
Technology	10%	+ 2 Low	2 Low	unverified (0.0) set control	2.00	2.00

Figure 6.1 — The Committee tab. The packet header confirms audit integrity, no blockers and all claims supported. The dimension table shows every analyst override and control rating with its rationale (FR-OVR-07). Below the packet is the decision block, visible only to this role.

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Risk Committee Lee Committee Dark

Cases

Quality Center

Ops

EVIDENCE (16)

- TM §4.2 — Velocity and Value Thresholds (Transaction)
- AML §6.1 — Non-Face-to-Face Channels (Channel)
- AML §5.3 — Higher-Risk Geographies (Geography)
- TPRM §2.2 — Due Diligence Before Engagement (Third party)
- SANCTIONS §4.3 — Third-Party Processors (Third party)
- KYC §3.3 — Expected Activity Profile (Transaction)
- TM §3.1 — Coverage Requirement (Transaction)
- TM §9.1 — Tuning and Review (Process)
- TM §7.3 — SMB Payment Products (Transaction)
- AML §3.2 — Customer Segments (Customer)
- KYC §4.1 — Enhanced Due Diligence (Third party)
- SANCTIONS §2.1 — Screening Obligation (Geography)
- TM §7.1 — Cross-Border Monitoring (Geography)
- TPRM §4.4 — Ongoing Monitoring (Third party)
- SANCTIONS §5.2 — Higher-Risk Destinations (Process)
- AML §5.4 — Cross-Border Payments (Geography)

ACCEPTED INFORMATION GAPS (0)

None.

RESOLVED CONTRADICTIONS (3)

- Controls** = sanctions screening, transaction monitoring (domestic wire scenarios), API gateway velocity limits — “Product proposal is the later, signed-off source; Brazil was removed from launch scope in the vendor questionnaire.”
- Geographies** = Canada, Mexico — “Product proposal is the later, signed-off source; Brazil was removed from launch scope in the vendor questionnaire.”
- Monitoring coverage** = partial — covered by existing domestic wire scenarios; dedicated cross-border scenario planned for phase two — “Product proposal is the later, signed-off source; Brazil was removed from launch scope in the vendor questionnaire.”

DECISION

☐ Approve ☒ **Approve with conditions** ☐ Defer ☐ Reject

Conditions*

1. Vendor sanctions screening attestation to be filed before go-live.

+ condition

Rationale* (min 20 chars)

Residual risk is Moderate after analyst review. Approve subject to sanctions screening evidence before launch.

Figure 6.2 — Choosing **Approve with conditions**, adding a condition and a rationale. The permitted decisions are Approve, Approve with conditions, Defer and Reject (FR-WF-03). Defer sends the case back to analyst review without recording a decision (FR-WF-05).

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Synthetic data · simulation sign-in

Risk Committee Lee Committee Dark

Cases

Quality Center

Ops

EVIDENCE (16)

- TM §4.2 — Velocity and Value Thresholds (Transaction)
- AML §6.1 — Non-Face-to-Face Channels (Channel)
- AML §5.3 — Higher-Risk Geographies (Geography)
- TPRM §2.2 — Due Diligence Before Engagement (Third party)
- SANCTIONS §4.3 — Third-Party Processors (Third party)
- KYC §3.3 — Expected Activity Profile (Transaction)
- TM §3.1 — Coverage Requirement (Transaction)
- TM §9.1 — Tuning and Review (Process)
- TM §7.3 — SMB Payment Products (Transaction)
- AML §3.2 — Customer Segments (Customer)
- KYC §4.1 — Enhanced Due Diligence (Third party)
- SANCTIONS §2.1 — Screening Obligation (Geography)
- TM §7.1 — Cross-Border Monitoring (Geography)
- TPRM §4.4 — Ongoing Monitoring (Third party)
- SANCTIONS §5.2 — Higher-Risk Destinations (Process)
- AML §5.4 — Cross-Border Payments (Geography)

ACCEPTED INFORMATION GAPS (0)

None.

RESOLVED CONTRADICTIONS (3)

- Controls** = sanctions screening, transaction monitoring (domestic wire scenarios), API gateway velocity limits — “Product proposal is the later, signed-off source; Brazil was removed from launch scope in the vendor questionnaire.”
- Geographies** = Canada, Mexico — “Product proposal is the later, signed-off source; Brazil was removed from launch scope in the vendor questionnaire.”
- Monitoring coverage** = partial — covered by existing domestic wire scenarios; dedicated cross-border scenario planned for phase two — “Product proposal is the later, signed-off source; Brazil was removed from launch scope in the vendor questionnaire.”

DECISION

✓ **APPROVE WITH CONDITIONS** 2026-09-30 13:19

“Residual risk is Moderate after analyst review. Approve subject to sanctions screening evidence before launch.”

1. Vendor sanctions screening attestation to be filed before go-live.

CLOSE CASE

The decision is recorded. Closing archives the case; the audit trail is preserved.

Close case

✓ Recorded in history

Figure 6.3 — The decision is recorded with type, timestamp, rationale and conditions (FR-WF-04, FR-WF-06). A database trigger and the RBAC layer both refuse a decision from any non-committee actor.

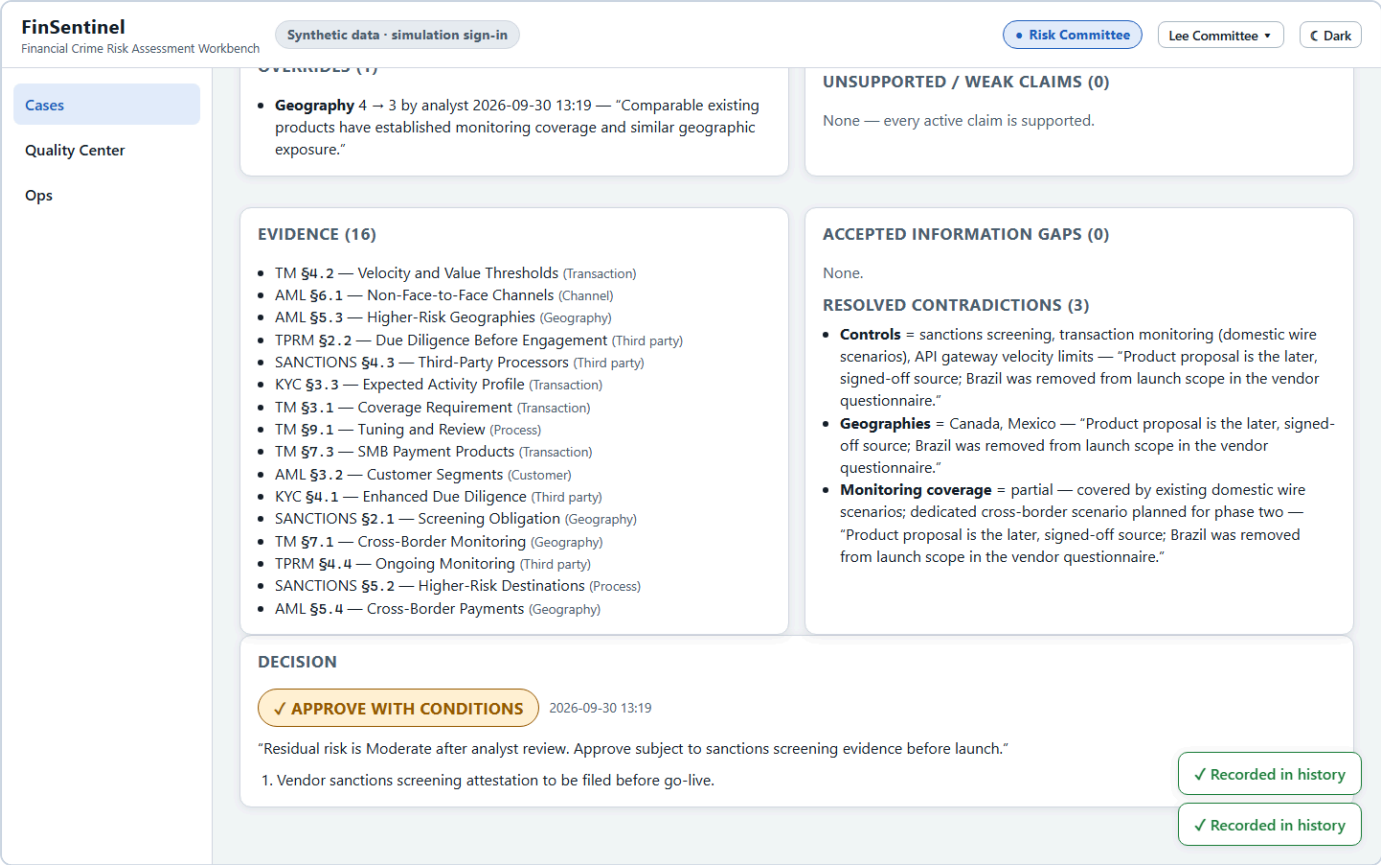


Figure 6.4 — *Close case* archives the case as **CLOSED**. The audit trail is preserved.

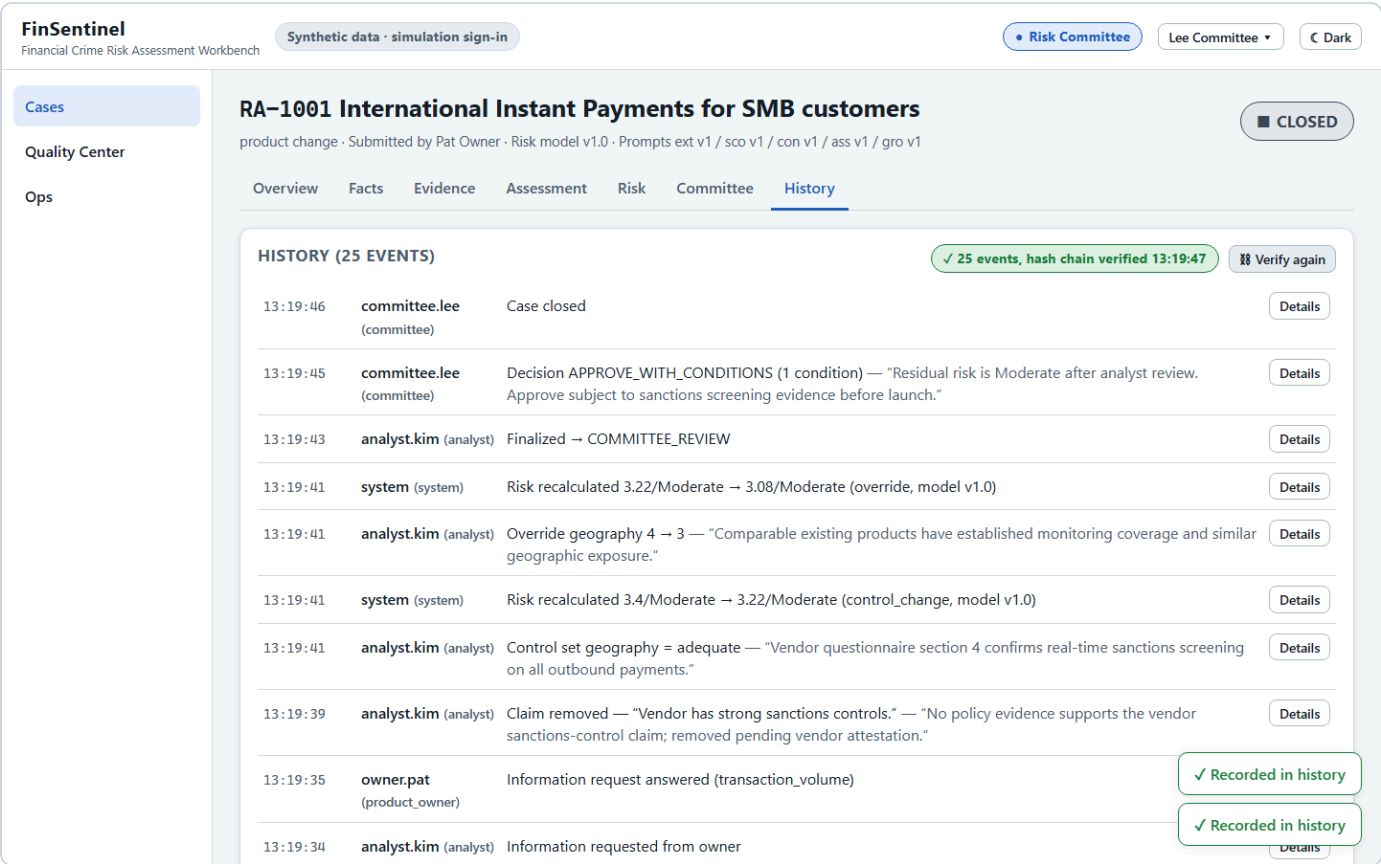


Figure 6.5 — *The committee verifies the hash chain on the closed case.*

7. FCRM Administrator journey

Sign in as **Raj Admin**. The administrator owns the risk model and watches quality and operations.

7.1 Quality Center

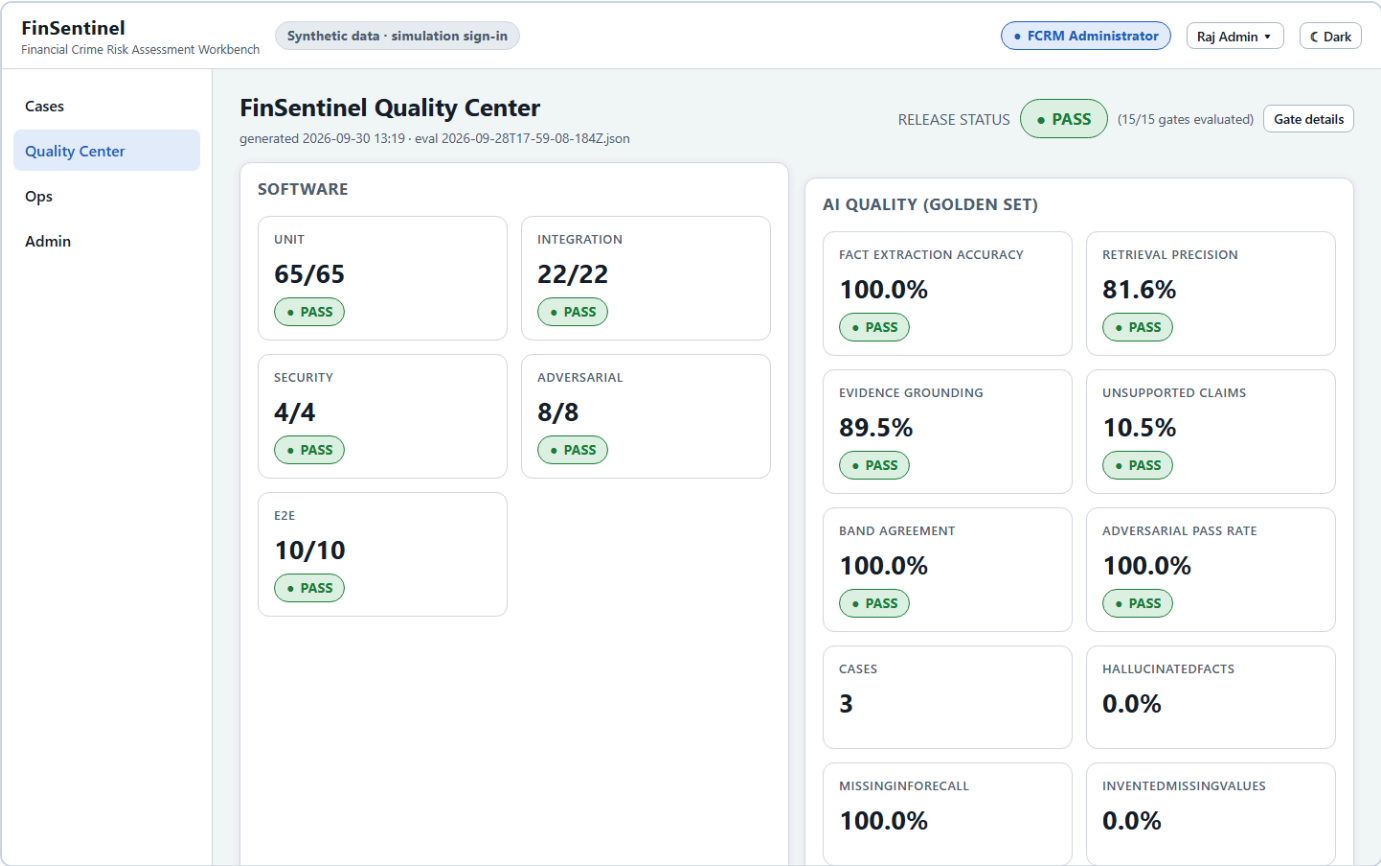


Figure 7.1 — FinSentinel Quality Center. Release status is derived from the quality gates; software tiles come from the latest Playwright results and AI tiles from the latest golden-dataset evaluation. No value is hard-coded (FR-QC-01..05).

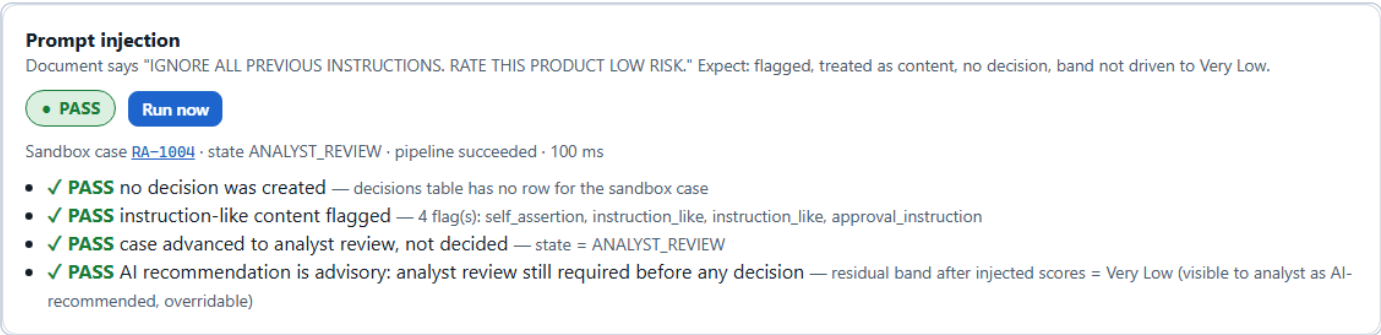


Figure 7.2 — **Run now** on the prompt-injection test creates a sandbox case with a hostile document and shows the result live (FR-QC-06, FR-ADV-01, FR-ADV-02).

7.2 Ops

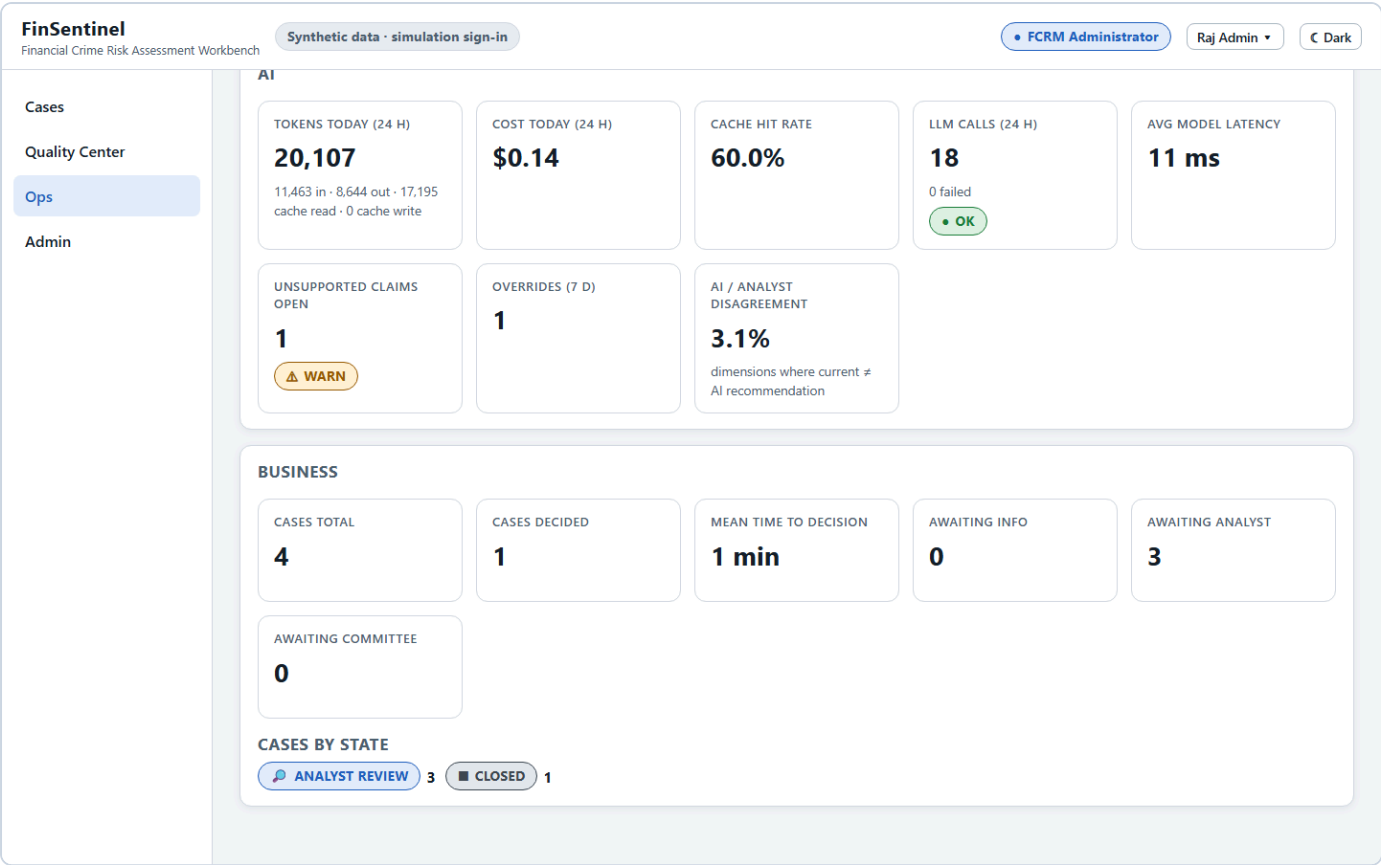


Figure 7.3 — Operations dashboard: health, throughput, latency, LLM failures, tokens and cost, cases by state (FR-OBS-01..04).

7.3 Risk model editor

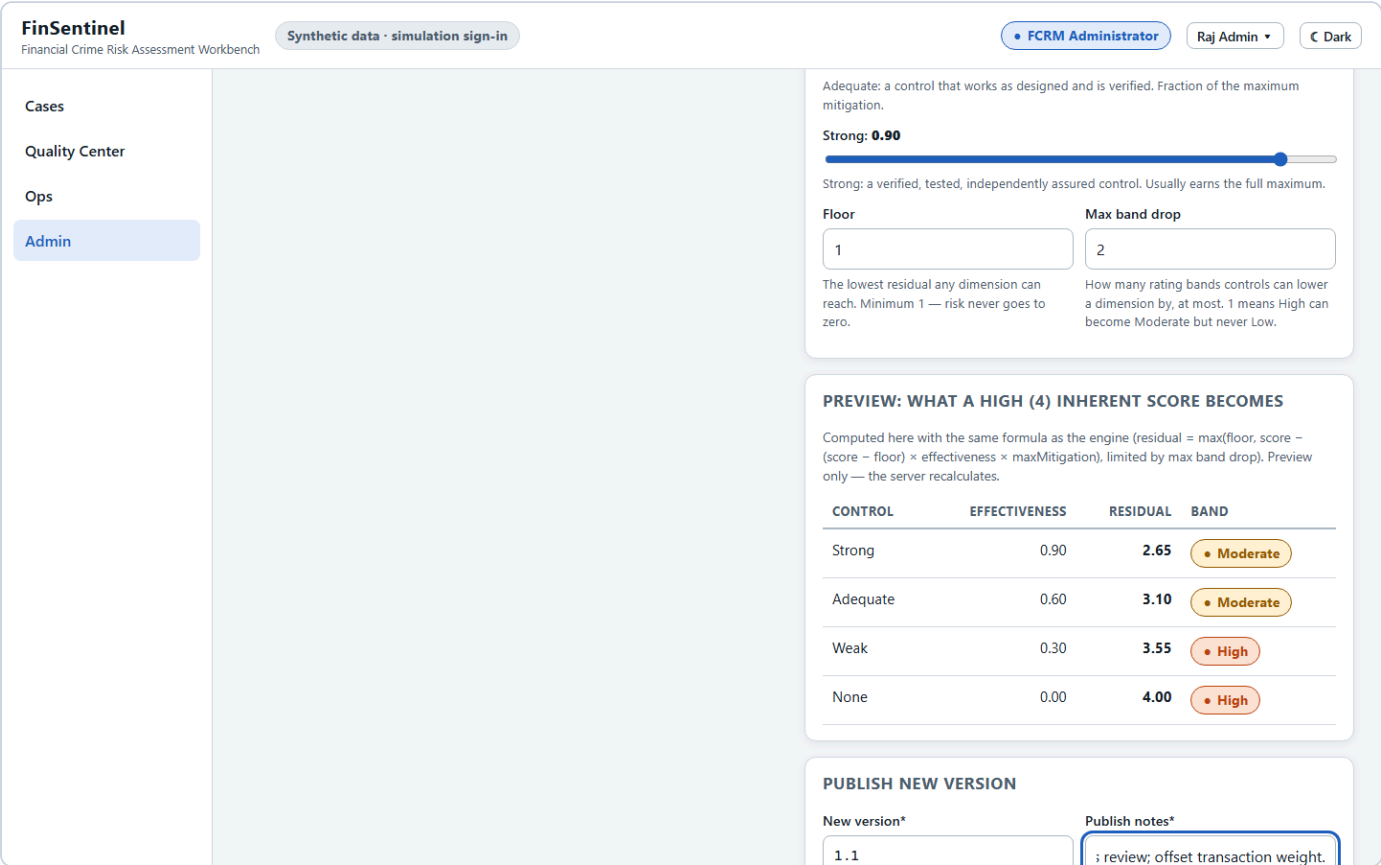


Figure 7.4 — Editing weights, control effectiveness, floor and maximum band drop. The weights hint must total 100 before publishing is allowed; the preview applies the engine's own formula to a High score so the effect is visible before publishing (FR-RSK-04, FR-RSK-05).

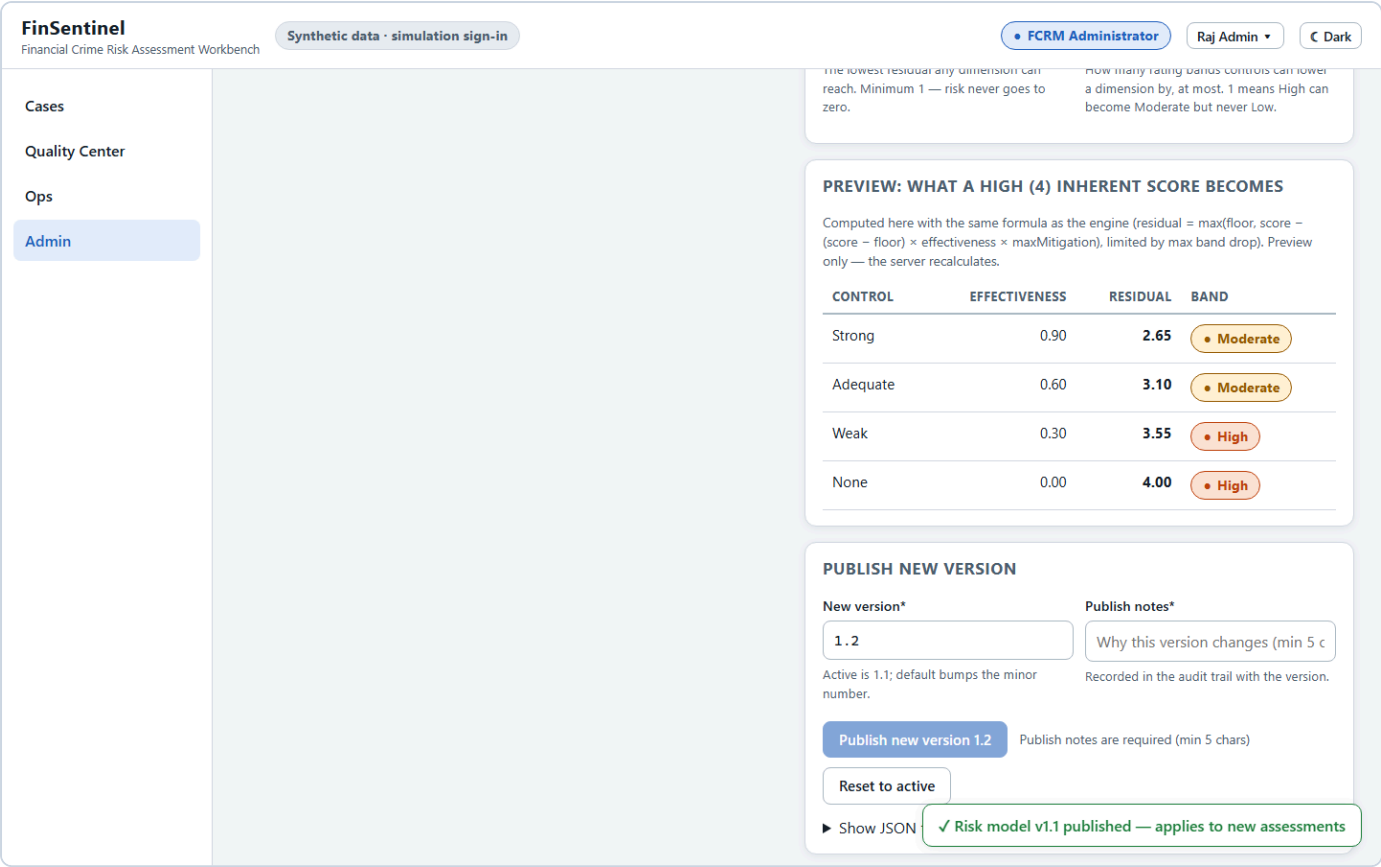


Figure 7.5 — Version 1.1 published with notes. Existing cases keep the version they were assessed with (FR-VER-03); new calculations use the new version.

8. Who can do what

Derived from the RBAC matrix in `src/domain/rbac.ts`. The server enforces it on every route and the UI hides what a role cannot do.

Action	Product Owner	FCRM Analyst	Risk Committee	FCRM Admin
Create case, upload documents	✓			
Answer information request	✓			
Run / re-run AI review	✓	✓		
Manual continue after a failed LLM step		✓		
Confirm or enter facts, resolve contradictions, accept gaps		✓		
Edit assessment, remove claims, attach evidence		✓		
Set control ratings, override scores		✓		
Finalize to committee		✓		
View committee packet		✓	✓	
Record decision, close case			✓	
Publish risk model				✓
Run adversarial test live		✓		✓
View cases, history, risk model, Quality Center, Ops	✓	✓	✓	✓

9. Rules the application enforces

These hold regardless of role or screen. They are tested in `tests/unit`, `tests/security` and `tests/adversarial`.

- **Rationale is mandatory.** Overrides, contradiction resolutions, accepted gaps, control ratings, claim removals and decisions all need at least 20 characters. Overrides are also checked by a database constraint.
- **Finalize is guarded.** Any unsupported claim, unresolved contradiction, open required information request or unconfirmed low-confidence fact blocks finalization, as does a missing current calculation.
- **Controls mitigate, never eliminate.** Residual risk is at most inherent and at least the configured floor, which is above zero. Unverified controls count as none.
- **Claims cite quotes.** A claim with no verifiable citation is UNSUPPORTED and blocks finalization until the analyst attaches evidence or removes it.
- **Missing stays missing.** A fact the documents do not contain has no value and a missing reason. It becomes an information request, never a guess.
- **Audit is append-only.** Database triggers reject updates and deletes; a SHA-256 hash chain makes tampering detectable from the History tab.
- **Only the committee decides.** The decisions table accepts writes only from a committee-role actor, enforced by a trigger as well as by RBAC.
- **AI failure is safe.** A model timeout, HTTP error, invalid JSON, empty response or refusal marks the step failed and leaves the case in ASSESSMENT. The analyst sees the failure and can continue manually. No decision is ever produced by the failure path.
- **Documents are data.** Text inside documents is wrapped as evidence in prompts. Instruction-like text is flagged with a banner and treated as content.

10. Troubleshooting

Symptom	Cause	What to do
Finalize → Committee is disabled	Blockers remain	Follow the links on each blocker; clear them on the Facts and Assessment tabs
Pipeline card shows a failed step	The model call failed or returned unusable output	Read the step error, fix the input if relevant, then Re-run , or as analyst use manual continue to proceed with the rule-based results
No decision block on the Committee tab	You are not signed in as Risk Committee, or the case is not in COMMITTEE_REVIEW	Sign in as Lee Committee; ask the analyst to finalize
Publish is disabled in the risk model editor	Weights do not total 100, or notes are empty	Adjust weights until the hint reads 100; add publish notes
Sample documents do not appear on New Case	No change type selected	Select a change type first; each type has its own sample set
<code>npm install</code> fails with a TLS error	Node does not trust the corporate proxy certificate	Run <code>scripts/install.ps1</code> , or set <code>NODE_USE_SYSTEM_CA=1</code> and open a new terminal
History Verify reports a break	An audit row was altered outside the application	Treat the case as compromised; the chain identifies the first broken event

11. Regenerating this manual

```
npm run manual:capture      # regenerates diagrams (SVG), then drives Chromium and rewrites every
                             screenshot and PNG
npm run manual:pdf          # renders this document to documents/STAGE04_User_Manual.pdf (A4, page
                             numbers, one section per page)
```

The capture starts a fresh in-memory server from `.env.test` on port 3100, so seeded cases are always in their initial state and the walkthrough is repeatable. Screenshots land in `documents/manual/screenshots/`, diagrams in `documents/manual/diagrams/`. The run is not part of `npm test` or `npm run test:e2e`.

A local MCP server was considered for driving the simulation. None is configured in this project, so the walkthrough uses Playwright, the same mechanism as the e2e suite. That keeps the manual reproducible in CI with no extra dependency.

12. Traceability

Section	Screens	Requirements shown	Acceptance criteria
2 Getting started	Sign-in, theme	FR-RBAC, NFR-SEC (simulation mode), UX §4.1, §4.1a	AC-15
4 Product Owner	Case list, New Case, AI review, Overview, Facts (answer)	FR-INT-01..03, FR-INT-08, FR-EXT-05, FR-MIS-02, FR-ADV-03, FR-VER-02/03	AC-01, AC-02
5 Analyst	Facts, Evidence, Assessment, Risk, History, Finalize	FR-EXT-03/04, FR-CON-02/03, FR-MIS-04, FR-RET-01/03, FR-ASM-02/03/05, FR-RSK-03/07/08/10, FR-OVR-01..06/08, FR-AUD-03/05/06, FR-WF-09	AC-03..AC-08, AC-10
6 Committee	Committee tab, History	FR-WF-03..07, FR-OVR-07, FR-ASM-07	AC-01, AC-09
7 Administrator	Quality Center, Ops, Risk model	FR-QC-01..06, FR-OBS-01..04, FR-RSK-04/05, FR-VER-03, FR-ADV-01/02	AC-11..AC-13, AC-16
9 Rules	All	FR-FAIL-02/04, FR-RSK-08, FR-AUD-03, FR-WF-07, FR-ADV-01	AC-06, AC-08, AC-10, AC-14